

Geometric Formulation for the Quasi-Steady Dynamics of a Flapping Wing UAV with Applications to Monarch Butterfly

Taeyoung Lee

February 17, 2022

Contents

1.	Flapping Wing UAV Model	1
1.1	Body	2
1.2	Wing	2
1.3	Wing Kinematics	5
1.4	Abdomen	5
2.	Quasi-Steady Aerodynamics	6
2.1	Wing Morphological Parameters	6
2.2	Blade-Element Theory	8
2.3	Translational Forces	9
2.4	Translational Forces for Small Body Velocities	10
2.5	Rotational Force	12
2.6	Rotational Force for Small Body Velocities	13
2.7	Numerical Example	14
3.	Flapping Wing UAV Dynamics	18
3.1	Lagrangian Mechanics on a Lie Group	18
3.2	Lagrangian Mechanics of Flapping Wing UAV	19
3.3	Prescribed Wing Kinematics and Abdomen Attitude	23
3.4	Prescribed Flapping/Deviation Angle and Abdomen Attitude	24
3.5	Prescribed Wing Kinematics and Abdomen/Body Attitude	26
3.6	Linearization of dynamics	26
3.7	Linearization with prescribed wing kinematics and abdomen/body attitude	30
4.	Dynamic Simulation for Monarch Butterfly	30
4.1	Monarch Morphological Parameters	30
4.2	Flight Characteristics of Monarch	33
4.3	Dynamic Simulation	34
4.4	Effects of Abdomen	36
4.5	Asymptotic Stability	36
5.	Floquet stability analysis	36
5.1	Periodic orbit generation	36
5.2	Floquet stability	39
6.	Control design	40
6.1	Stability of controlled trajectory	41

1. Flapping Wing UAV Model

Consider a flapping wing UAV that is composed of a head, a thorax, an abdomen, and two wings attached to the thorax. Here, we assume that the head and the thorax are coagulated into a single rigid body, which is referred to as

body. We plan to study the motion of the abdomen relative to the thorax in the future. Also, we do not distinguish hindwings from forewings.

Define an inertial frame $\mathcal{F}_I = \{\mathbf{i}_x, \mathbf{i}_y, \mathbf{i}_z\}$, where the third axis points downward, and the first two axes span the horizontal plane. This is compatible to the NED (north-east-down) frame common in flight dynamics.

1.1 Body

Define the body-fixed frame $\mathcal{F}_B = \{\mathbf{b}_x, \mathbf{b}_y, \mathbf{b}_z\}$, whose origin is located at its mass center of the body. Following the common convention in flight dynamics, the first axis points toward the head, the second axis point toward the right wing, and the third axis points toward the ventral (belly) side.

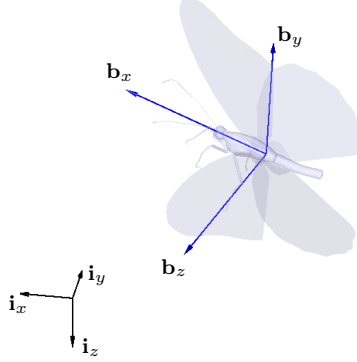


Figure 1: The inertial frame $\mathcal{F}_I = \{\mathbf{i}_x, \mathbf{i}_y, \mathbf{i}_z\}$ (black) and the body-fixed frame $\mathcal{F}_B = \{\mathbf{b}_x, \mathbf{b}_y, \mathbf{b}_z\}$ (blue)

The location of the mass center is given by $x \in \mathbb{R}^3$ in $\mathcal{F}_I$. The attitude of the body is described by $R \in \text{SO}(3) = \{R \in \mathbb{R}^{3 \times 3} \mid R^T R = I_{3 \times 3}, \det[R] = 1\}$, which represents the linear transformation of a representation of a vector from $\mathcal{F}_B$ to $\mathcal{F}_I$. The attitude kinematics is given by

$$\dot{R} = R\hat{\Omega}, \quad (1)$$

where $\Omega \in \mathbb{R}^3$ is the angular velocity resolved in $\mathcal{F}_B$. In the above equation, the *hat map* $\wedge : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ is defined such that $\hat{x}y = x \times y$ for any $x, y \in \mathbb{R}^3$, or equivalently

$$\hat{x} = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}, \quad (2)$$

for $x = (x_1, x_2, x_3) \in \mathbb{R}^3$. The inverse of the hat map is denoted by the *vee map*, $\vee : \mathfrak{so}(3) \rightarrow \mathbb{R}^3$.

1.2 Wing

Right Wing Let $\mathcal{F}_R = \{\mathbf{r}_x, \mathbf{r}_y, \mathbf{r}_z\}$ be the frame fixed to the right wing. Its origin is located at the joint of the right wing, or the wing root where the right wing is attached to the thorax. The vector from the origin of $\mathcal{F}_B$ to the origin of $\mathcal{F}_R$ is defined as $\mu_R \in \mathbb{R}^3$. As it is resolved in $\mathcal{F}_B$ and the body is rigid, $\dot{\mu}_R = 0$. The first two axes of $\mathcal{F}_R$ span the plane of the wing, where the first axis points toward the leading edge and the second axis points toward the wing tip. Consequently, the third axis is normal to the wing plane, and it points toward the ventral side when there is no rotation of the right wing.

Next, we introduce the stroke frame $\mathcal{F}_S = \{\mathbf{s}_x, \mathbf{s}_y, \mathbf{s}_z\}$, which is constructed by translating the origin of $\mathcal{F}_B$ to the center of the left wing root and the right wing root, and rotating it about $\mathbf{b}_y$ by $\beta \in [-\pi, \pi)$. The y - z plane of $\mathcal{F}_S$ is referred to as the stroke plane.

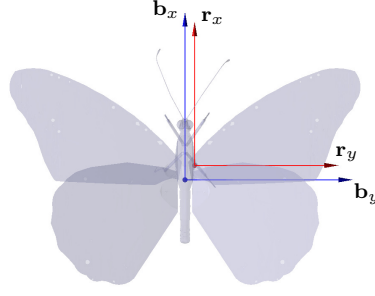


Figure 2: The body-fixed frame $\mathcal{F}_B = \{\mathbf{b}_x, \mathbf{b}_y, \mathbf{b}_z\}$ (blue), the right wing frame $\mathcal{F}_R = \{\mathbf{r}_x, \mathbf{r}_y, \mathbf{r}_z\}$ (red)

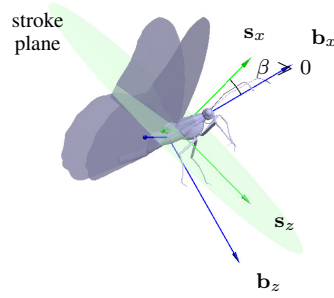


Figure 3: The body-fixed frame $\mathcal{F}_B = \{\mathbf{b}_x, \mathbf{b}_y, \mathbf{b}_z\}$ (blue), the stroke frame $\mathcal{F}_S = \{\mathbf{s}_x, \mathbf{s}_y, \mathbf{s}_z\}$ (green)

The angle β can be considered as the angle between $\mathbf{b}_x$, and $\mathbf{s}_x$ that is normal to the stroke plane. Or equivalently, it is the angle between y - z plane of $\mathcal{F}_B$ and the stroke plane. It is positive when $\mathbf{s}_x \cdot \mathbf{b}_z < 0$. This angle is often defined as the angle between the ground, horizontal plane and the stroke plane, considering hovering flights with a fixed attitude of the body. Here, it is defined relative to $\mathcal{F}_B$ as the flapping motion is inherently relative to the body.

The motion of the right wing relative to $\mathcal{F}_S$ is described by 1–3–2 Euler angles $(\phi_R, \psi_R, \theta_R)$. Consequently, the orientation of the right wing relative to $\mathcal{F}_B$, namely $Q_R \in \text{SO}(3)$ is described by

$$Q_R(t) = \exp(\beta \hat{e}_2) \exp(\phi_R(t) \hat{e}_1) \exp(-\psi_R(t) \hat{e}_3) \exp(\theta_R(t) \hat{e}_2), \quad (3)$$

which is the linear transformation of a representation of a vector from $\mathcal{F}_R$ to $\mathcal{F}_B$. The Euler-angles are defined as

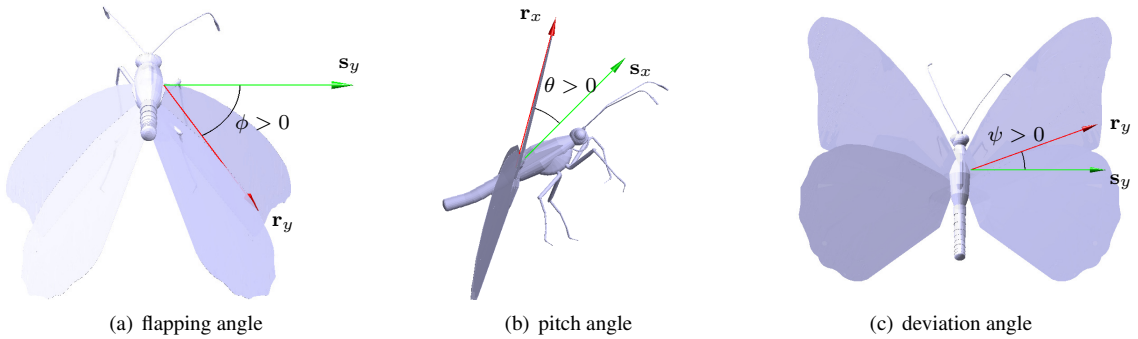


Figure 4: Wing configuration when only one of (ϕ, θ, ψ) is non-zero

$\phi_R \in [-\pi, \pi)$ the flapping angle, which is positive when the wing is in the ventral side, i.e., $\dot{\phi} > 0$ corresponds to downstroke and $\dot{\phi} < 0$ corresponds to upstroke (this is also referred to as the stroke angle or the sweep angle in other papers)

$\theta_R \in [-\pi, \pi)$ the pitch angle about the axis from the wing root to the wing tip, which is positive when the leading edge of the wing is rotated toward the dorsal side (this is also referred to as the feathering angle or the rotation angle in other papers)

$\psi_R \in [-\pi, \pi)$ the deviation angle that governs the motion of the wing tip out of the stroke plane, which is positive when the wing tip is rotated toward the head

The time-derivative of Q_R is given by

$$\dot{Q}_R = Q_R \hat{\Omega}_R, \quad (4)$$

where $\Omega_R \in \mathbb{R}^3$ is the angular velocity of the right wing relative to $\mathcal{F}_B$ resolved in $\mathcal{F}_R$. One can show that the angular velocity of the right wing is obtained from the time-derivatives of the Euler-angles as

$$\Omega_R = \begin{bmatrix} \cos \psi_R \cos \theta_R & 0 & \sin \theta_R \\ \sin \psi_R & 1 & 0 \\ \cos \psi_R \sin \theta_R & 0 & -\cos \theta_R \end{bmatrix} \begin{bmatrix} \dot{\phi}_R \\ \dot{\theta}_R \\ \dot{\psi}_R \end{bmatrix}. \quad (5)$$

The determinant of the above 3×3 matrix is $-\cos \psi_R$. Consequently, it is invertible when the deviation angle satisfies $\psi_R \neq \pm \frac{\pi}{2}$. This is not restrictive as the deviation angle is small in general.

Left Wing Similarly, let $\mathcal{F}_L$ be the frame fixed to the left wing. It can be obtained by translating $\mathcal{F}_R$ to the root of the left wing without any rotation. More specifically, its origin is located at the joint of the left wing, where the left wing is attached to the thorax. The first two axes span the plane of the wing, where the first axis points toward the leading edge and the second axis points toward the right wing, opposite to the left wing tip. Consequently, the third axis is normal to the wing plane, and it points toward the ventral side when there is no rotation of the left wing.

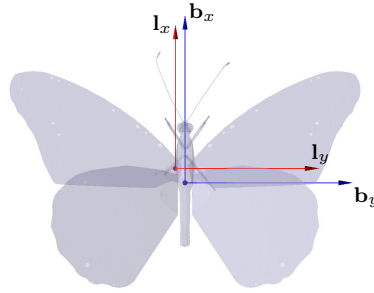


Figure 5: The body-fixed frame $\mathcal{F}_B = \{\mathbf{b}_x, \mathbf{b}_y, \mathbf{b}_z\}$ (blue), the left wing frame $\mathcal{F}_L = \{\mathbf{l}_x, \mathbf{l}_y, \mathbf{l}_z\}$ (red)

Similar as the right wing, the motion of the left wing relative to the thorax is described by $Q_L \in \text{SO}(3)$, which is described by successive rotations as

$$Q_L(t) = \exp(\beta \hat{e}_2) \exp(-\phi_L(t) \hat{e}_1) \exp(\psi_L(t) \hat{e}_3) \exp(\theta_L(t) \hat{e}_2), \quad (6)$$

where the definition of the Euler-angles $(\phi_L, \theta_L, \psi_L)$ are consistent with those of the right wing. The wing kinematics become symmetric when $(\phi_R, \theta_R, \psi_R) = (\phi_L, \theta_L, \psi_L)$.

The time-derivative of Q_L is given by

$$\dot{Q}_L = Q_L \hat{\Omega}_L, \quad (7)$$

where $\Omega_L \in \mathbb{R}^3$ is the angular velocity of the left wing relative to $\mathcal{F}_B$ resolved in $\mathcal{F}_L$. Also, we have

$$\Omega_L = \begin{bmatrix} -\cos \psi_L \cos \theta_L & 0 & -\sin \theta_L \\ \sin \psi_L & 1 & 0 \\ -\cos \psi_L \sin \theta_L & 0 & \cos \theta_L \end{bmatrix} \begin{bmatrix} \dot{\phi}_L \\ \dot{\theta}_L \\ \dot{\psi}_L \end{bmatrix}. \quad (8)$$

1.3 Wing Kinematics

We consider the wing kinematics model presented in [1]. Let $f \in \mathbb{R}$ be the frequency of the flapping in Hz. Thus, the period of the flapping is $T = \frac{1}{f}$.

The flapping angle is given by a smoothed triangular waveform,

$$\phi(t) = \frac{\phi_m}{\sin^{-1} \phi_K} \sin^{-1}(\phi_K \cos(2\pi f t)) + \phi_0, \quad (9)$$

where $\phi_m \in \mathbb{R}$ is the amplitude of flapping, and $\phi_0 \in \mathbb{R}$ is the offset. The parameter $0 < \phi_K \leq 1$ determines the shape of flapping waveform (sinusoidal when $\phi_K \rightarrow 0$; triangular when $\phi_K \rightarrow 1$). The initial pitch angle is $\phi(0) = \phi_m + \phi_0$. During $0 \leq t \leq \frac{T}{2}$, the pitch angle decreases to $\phi(\frac{T}{2}) = -\phi_m + \phi_0$. Therefore, it represents upstroke when $0 \leq t \leq \frac{T}{2}$, and it represents downstroke when $\frac{T}{2} \leq t \leq T$.

In [1], the inner term is given by $\sin(2\pi f t)$, instead of $\cos(2\pi f t)$ as in (9). By following the convention of [2], we use the cosine. This is to ensure that in the subsequent expression of the pitch angle, the pitch angle reversal happens at the end of the downstroke or at the end of upstroke when there is no phase offset ($\theta_a = 0$).

The pitch angle is given by a hyperbolic function,

$$\theta(t) = \frac{\theta_m}{\tanh \theta_C} \tanh(\theta_C \sin(2\pi f t + \theta_a)) + \theta_0, \quad (10)$$

where $\theta_m \in \mathbb{R}$ is the amplitude of pitching, and $\theta_0 \in \mathbb{R}$ is the offset. The parameter $\theta_C \in (0, \infty)$ determines the waveform (sinusoidal when $\theta_C \rightarrow 0$; step function when $\theta_C \rightarrow \infty$). The value of θ_C is related to the duration of the wing pitch reversal. The parameter $\theta_a \in (-\pi, \pi)$ describes the offset in phase (advance rotation for $\theta_a > 0$; symmetric rotation for $\theta_a = 0$; delayed rotation for $\theta_a < 0$).

Finally, the deviation angle is given by a sinusoidal oscillation,

$$\psi(t) = \psi_m \cos(2\pi \psi_N f t + \psi_a) + \psi_0, \quad (11)$$

where $\psi_m \in \mathbb{R}$ is the amplitude of the deviation, and $\psi_0 \in \mathbb{R}$ is the offset. The parameter ψ_N is either 1 or 2: $\psi_N = 1$ corresponds to one oscillation per a flapping period, and $\psi_N = 2$ corresponds to a figure-eight motion. The parameter $\psi_a \in (-\pi, \pi)$ describes the offset in phase. See Figure 6 for the trajectories of these angles for selected parameters.

1.4 Abdomen

Consider an abdomen that is assumed to be a rigid body attached to the thorax via a spherical joint. Let $\mathcal{F}_A = \{\mathbf{a}_x, \mathbf{a}_y, \mathbf{a}_z\}$ be the frame fixed to the abdomen. Its origin is located at the mass center of the abdomen, and its orientation is identical to $\mathcal{F}_B$ when there is no rotation relative to the body. The vector from the origin of $\mathcal{F}_B$ to the joint connecting the thorax and the abdomen is $\mu_A \in \mathbb{R}^3$. As it is resolved in $\mathcal{F}_B$ and the body is rigid, we have $\dot{\mu}_A = 0$.

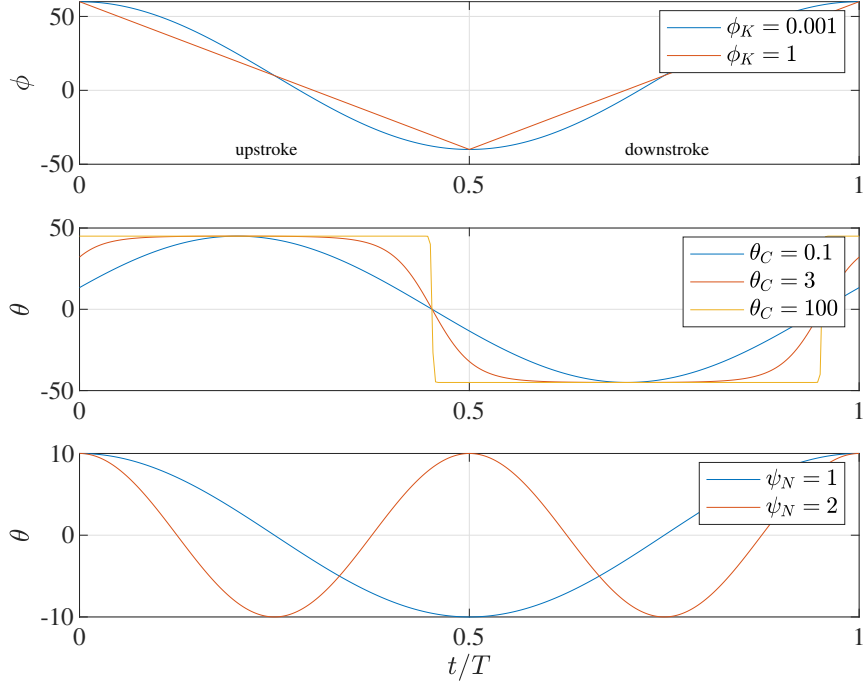


Figure 6: Wing kinematics when $\phi_m = 50^\circ$, $\phi_0 = 10^\circ$, $\theta_m = 45^\circ$, $\theta_0 = 0$, $\theta_a = 0.3$, $\phi_m = 10^\circ$, $\phi_0 = \phi_a = 0$

The motion of the abdomen relative to the body is described by $Q_A \in \text{SO}(3)$, which is the linear transformation of the representation of a vector from $\mathcal{F}_A$ to $\mathcal{F}_B$. Its time-derivative is given by

$$\dot{Q}_A = Q_A \hat{\Omega}_A, \quad (12)$$

where $\Omega_A \in \mathbb{R}^3$ is the angular velocity of the abdomen relative to the body, resolved in $\mathcal{F}_A$.

2. Quasi-Steady Aerodynamics

The quasi-steady assumption implies that the aerodynamic force and moment generated by the flapping wing are equivalent to those for steady motion at the same instantaneous velocity and the angle of attack.

2.1 Wing Morphological Parameters

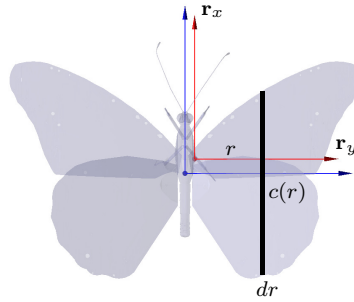


Figure 7: Infinitesimal wing segment

We first present several morphological parameters of the wing as formulated in to [3], after making a few changes in the notation. More specifically, in [3], the symbol $\wedge$ is used to denote normalized variables. However, the same symbol is used for the hat map in (2). Instead, we use the symbol $\tilde{\cdot}$ for the normalized variables to avoid conflict with (2). Also, the morphological parameters are defined for a single wing as the aerodynamic force of the left wing will be distinguished from the right wing for non-symmetric flapping.

Let dr be the infinitesimal wing segment located distanced at r from the wing root. Let its chord be defined as $c(r)$. The wing segment is parallel to $\mathbf{r}_x$ and the distance is measured along $\mathbf{r}_y$. Assume that the aerodynamic center of the segment is along $\mathbf{r}_y$, i.e., the resultant force generated by the wing segment is located along $\mathbf{r}_y$ with zero moment. Consequently, it is located at re_2 when resolved in $\mathcal{F}_R$.

The area of the right wing is given by

$$S = \int_0^l c(r)dr, \quad (13)$$

where $l > 0$ is the wing span, i.e., the distance between the wing root and the wing tip measure along $\mathbf{r}_y$. The non-dimensional aspect ratio is

$$\mathcal{R} = \frac{l^2}{S}. \quad (14)$$

The mean chord is the area divided by the span, i.e., $\bar{c} = \frac{S}{l}$, and the normalized wing chord is

$$\tilde{c} = \frac{c}{\bar{c}} = \frac{cl}{S}. \quad (15)$$

Also, the non-dimensional radius is defined as

$$\tilde{r} = \frac{r}{l}. \quad (16)$$

The k -th moment of wing area is defined as

$$S_k = \int_0^l r^k c(r)dr = Sl^k \int_0^1 \tilde{r}^k \tilde{c}d\tilde{r}, \quad (17)$$

satisfying $S_0 = S$. The non-dimensional radius of the k -th moment of wing area is defined as

$$\tilde{r}_k = \left(\frac{S_k}{Sl^k} \right)^{\frac{1}{k}} = \left(\frac{1}{Sl^k} \int_0^l r^k c(r)dr \right)^{\frac{1}{k}} = \left(\int_0^1 \tilde{r}^k \tilde{c}d\tilde{r} \right)^{\frac{1}{k}}, \quad (18)$$

such that $S_k = (\tilde{r}_k l)^k S$, i.e., if all of the wing area is located at a distance $\tilde{r}_k l$, the k -th moment of area is equal to S_k .

Next, when a wing is accelerated, it causes the motion of the surrounding air. The inertia of the wing is increased by the mass of air that is accelerated, and therefore there is an increase in the wing mass. The normalized virtual mass is defined as

$$\tilde{v} = \int_0^1 \tilde{c}^2 d\tilde{r}. \quad (19)$$

And the corresponding normalized radius of the k -th moment of virtual mass is

$$\tilde{r}_k(v) = \left(\frac{1}{\tilde{v}} \int_0^1 \tilde{c}^2 \tilde{r}^k d\tilde{r} \right)^{\frac{1}{k}}. \quad (20)$$

Later, we need to compute the following quantity:

$$\int_0^l r c^2(r) dr = l^2 \bar{c}^2 \int_0^1 \tilde{r} \tilde{c}^2 d\tilde{r} = l^2 \bar{c}^2 \tilde{v} \tilde{r}_1(v), \quad (21)$$

$$\int_0^l r^2 c^2(r) dr = l^3 \bar{c}^2 \int_0^1 \tilde{r}^2 \tilde{c}^2 d\tilde{r} = l^2 \bar{c}^2 \tilde{v} \tilde{r}_2^2(v). \quad (22)$$

2.2 Blade-Element Theory

Angle of Attack of Right Wing When resolved in the inertial frame, the root of the right wing is located at $x + R\mu_R$. Thus, the aerodynamic center of the chord at the distance r from the wing root is

$$x + R\mu_R + RQ_R r e_2.$$

Therefore, its velocity in $\mathcal{F}_I$ is

$$\dot{x} + R\hat{\Omega}\mu_R + R\hat{\Omega}Q_R r e_2 + RQ_R\hat{\Omega} r e_2,$$

which is transformed to $\mathcal{F}_R$ by left-multiplying $Q_R^T R^T$ as

$$\begin{aligned} Q_R^T R^T \dot{x} + Q_R^T \hat{\Omega} \mu_R + Q_R^T \hat{\Omega} Q_R r e_2 + \hat{\Omega} r e_2 \\ = Q_R^T (R^T \dot{x} + \Omega \times \mu_R) + r(Q_R \Omega + \Omega_R) \times e_2, \end{aligned}$$

where the first term corresponds to the velocity of the wing root and the second term corresponds to the velocity of the aerodynamic center relative to the wing root. Both are resolved in $\mathcal{F}_R$.

Assume that there is a uniform wind with the velocity $v_{\text{wind}} \in \mathbb{R}^3$ resolved in $\mathcal{F}_I$. The velocity of the aerodynamic center of $c(r)$ relative to the wind is given by

$$Q_R^T (R^T (\dot{x} - v_{\text{wind}}) + \Omega \times \mu_R) + r(Q_R \Omega + \Omega_R) \times e_2,$$

According to the *blade-element theory* [3], the aerodynamic force generated by the infinitesimal chord is independent of the span-wise velocity component, i.e., the component of the above expression along $\mathbf{r}_y$ is irrelevant. Therefore, we project the above velocity to the $\mathbf{r}_x$ - $\mathbf{r}_z$ plane as follows.

$$U_R(r) = (I_{3 \times 3} - e_2 e_2^T) Q_R^T (R^T (\dot{x} - v_{\text{wind}}) + \Omega \times \mu_R) + r(Q_R \Omega + \Omega_R) \times e_2. \quad (23)$$

where $I_{3 \times 3} - e_2 e_2^T = \text{diag}[1, 0, 1] \in \mathbb{R}^{3 \times 3}$ corresponds to the projection operator. The second term is not affected by the projection as it is already normal to e_2 .

The angle of attack $\alpha_R(r)$ is the angle between the chord line $\pm e_1$ and the above velocity $U_R(r)$. Assuming that the chord is a thin blade, when the angle is greater than $\frac{\pi}{2}$, we flip the leading edge and the trailing edge of the chord. This ensures that $\alpha_R(r) \in [0, \frac{\pi}{2}]$ always. More explicitly,

$$\alpha_R(r) = \cos^{-1} \left(\frac{|e_1^T U_R(r)|}{\|U_R(r)\|} \right). \quad (24)$$

The above expression has a singularity when $U(r) = 0$. But, the value of $\alpha(r)$ does not matter when $U(r) = 0$, as the resulting aerodynamic force and moment will vanish. Or equivalently, it can be written as

$$\alpha_R(r) = \sin^{-1} \left(\frac{|e_3^T U_R(r)|}{\|U_R(r)\|} \right). \quad (25)$$

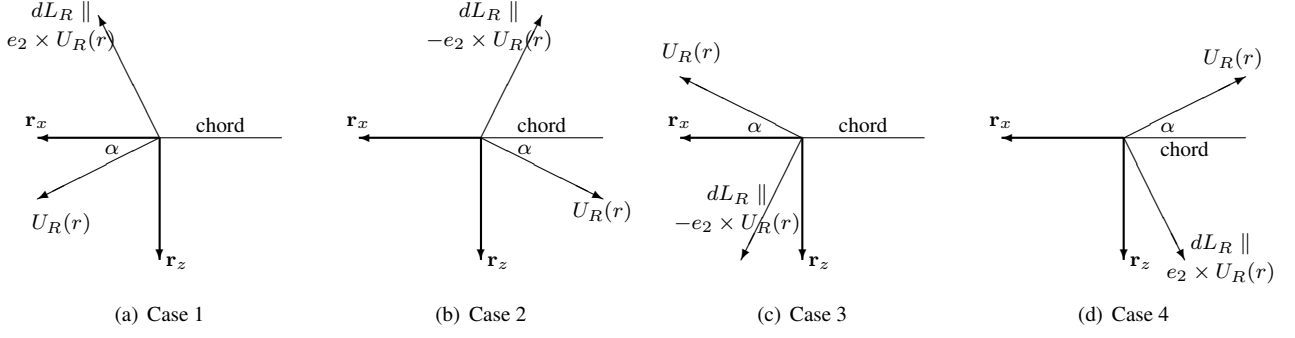


Figure 8: Direction of the velocity and the lift in the $\mathbf{r}_x$ - $\mathbf{r}_z$ plane of the right wing

Angle of Attack of Left Wing For the left wing, (23) is changed into

$$U_L(r) = (I_{3 \times 3} - e_2 e_2^T) Q_L^T (R^T (\dot{x} - v_{\text{wind}}) + \Omega \times \mu_L) - r(Q_L \Omega + \Omega_L) \times e_2. \quad (26)$$

The angle of attack is given by

$$\alpha_L(r) = \cos^{-1} \left(\frac{|e_1^T U_L(r)|}{\|U_L(r)\|} \right) = \sin^{-1} \left(\frac{|e_3^T U_L(r)|}{\|U_L(r)\|} \right). \quad (27)$$

2.3 Translational Forces

Right Wing Here we find the expression of the lift and the drag. The magnitude of the lift generated by the infinitesimal wing segment $c(r)$ is

$$\frac{1}{2} \rho \|U_R(r)\|^2 C_L(\alpha_R(r)) c(r) dr,$$

where $\rho \in \mathbb{R}$ is the atmospheric density, and $C_L \in \mathbb{R}$ is the lift coefficient given as a function of the angle of attack.

The direction of the lift is normal to both of the velocity $U_R(r)$ and the wing span-wise direction e_2 . As such, the direction of the lift is along $\pm e_2 \times U_R(r)$ in $\mathcal{F}_R$. The ambiguity of the sign can be resolved by the fact that when the $\mathbf{r}_x$ - $\mathbf{r}_z$ plane is divided by the chord line $\mathbf{r}_x$, the velocity vector $U(r)$ and the lift vector occupy the opposite side with each other.

More specifically, consider the four cases illustrated in Figure 8. These show that the direction of lift is $e_2 \times U_R(r)$ when $e_1^T U_R(r)$ and $e_3^T U_R(r)$ have the same sign (Case 1 and Case 4), and it is $-e_2 \times U_R(r)$ when they have the opposite sign (Case 2 and Case 3). Therefore, the infinitesimal lift vector resolved in $\mathcal{F}_R$ is given by

$$\begin{aligned} dL_R(r) &= \frac{1}{2} \rho U_R^2(r) C_L(\alpha(r)) c(r) \text{sgn}(e_1^T U_R(r) e_3^T U_R(r)) \frac{e_2 \times U_R(r)}{\|e_2 \times U_R(r)\|} dr \\ &= \frac{1}{2} \rho C_L(\alpha(r)) c(r) \text{sgn}(e_1^T U_R(r) e_3^T U_R(r)) (e_2 \times U_R(r)) \|U_R(r)\| dr, \end{aligned} \quad (28)$$

where the second equality is obtained by $\|U_R(r)\| = \|e_2 \times U_R(r)\|$.

The total lift of the right wing is obtained by integrating above span-wise for $r \in [0, l]$.

$$L_R = \int_0^l dL_R(r), \quad (29)$$

which is resolved in $\mathcal{F}_R$.

Next, the drag is always opposite to $U_R(r)$. Similar with the above expressions,

$$dD_R(r) = -\frac{1}{2} \rho C_D(\alpha(r)) c(r) \|U_R(r)\| U_R(r) dr. \quad (30)$$

The total drag is obtained by integrating above span-wise for $r \in [0, l]$ as

$$D_R = \int_0^l dD_R(r), \quad (31)$$

which is resolved in $\mathcal{F}_R$.

The lift and the drag also generate the moment about the wing root. When resolved in $\mathcal{F}_R$, it is given by

$$M_R = \int_0^l r e_2 \times (dL_R + dD_R). \quad (32)$$

Left Wing Similarly, the infinitesimal lift vector of the left wing resolved in $\mathcal{F}_L$ is given by

$$dL_L(r) = \frac{1}{2} \rho C_L(\alpha(r)) c(r) \text{sgn}(e_1^T U_L(r) e_3^T U_L(r)) (e_2 \times U_L(r)) \|U_L(r)\| dr, \quad (33)$$

which is integrated to obtain

$$L_L = \int_0^l dL_L(r). \quad (34)$$

Also, the infinitesimal drag of the left wing is

$$dD_L(r) = -\frac{1}{2} \rho C_D(\alpha(r)) c(r) \|U_L(r)\| U_L(r) dr, \quad (35)$$

and

$$D_L = \int_0^l dD_L(r), \quad (36)$$

which is resolved in $\mathcal{F}_R$.

The lift and the drag also generate the moment about the wing root as

$$M_L = \int_0^l -r e_2 \times (dL_L + dD_L). \quad (37)$$

2.4 Translational Forces for Small Body Velocities

This above expression requires span-wise integration at every instance to compute the lift and the drag. Here we present an explicit expression for a special case when the linear velocity and the angular velocity of the body and the wind velocity are small. i.e., $\|\dot{x}\|, \|\Omega\|, \|v_{\text{wind}}\| \ll 1$.

Right Wing Under these assumptions, the first part of (23) vanishes so that the velocity reduces to

$$U_R(r) = r \Omega_R \times e_2,$$

which is directly proportional to r . In (24), this eliminates the dependency of α on r as follows.

$$\alpha = \cos^{-1}\left(\frac{|e_1^T(\Omega_R \times e_2)|}{\|\Omega_R \times e_2\|}\right) = \cos^{-1}\left(\frac{|e_3^T \Omega_R|}{\|\Omega_R \times e_2\|}\right).$$

In other words, the angle of attack is identical for all the infinitesimal chord in the right wing. Also, from (28),

$$\begin{aligned} dL_R(r) &= \frac{1}{2}\rho C_L(\alpha)c(r)\text{sgn}(e_1^T(\Omega_R \times e_2)e_3^T(\Omega_R \times e_2))(e_2 \times (\Omega_R \times e_2))\|\Omega_R \times e_2\|r^2dr \\ &= \frac{1}{2}\rho C_L(\alpha)c(r)\text{sgn}(-e_3^T\Omega_R e_1^T\Omega_R)(e_2 \times (\Omega_R \times e_2))\|\Omega_R \times e_2\|r^2dr \\ &= \frac{1}{2}\rho C_L(\alpha)c(r)\text{sgn}(e_3^T\Omega_R e_1^T\Omega_R)\hat{e}_2^2\Omega_R\|\Omega_R \times e_2\|r^2dr. \end{aligned}$$

Consequently, the lift of the right wing is given by

$$L_R = \frac{1}{2}\rho C_L(\alpha)\text{sgn}(e_3^T\Omega_R e_1^T\Omega_R)\hat{e}_2^2\Omega_R\|\Omega_R \times e_2\| \int_0^l c(r)r^2dr,$$

and using (18)

$$L_R = \frac{1}{2}\rho C_L(\alpha)\text{sgn}(e_3^T\Omega_R e_1^T\Omega_R)\hat{e}_2^2\Omega_R\|\Omega_R \times e_2\|l^2\tilde{r}_2^2S,$$

which can be evaluated by Ω_R without integration over the wing span.

Or, equivalently, define the velocity of the chord at $r = \tilde{r}_2l$ as

$$U_R = U_R(\tilde{r}_2l) = \tilde{r}_2l\Omega_R \times e_2. \quad (38)$$

The angle of attack can be rewritten as

$$\alpha_R = \cos^{-1}\left(\frac{|e_1^TU_R|}{\|U_R\|}\right) = \sin^{-1}\left(\frac{|e_3^TU_R|}{\|U_R\|}\right). \quad (39)$$

Thus, the lift of the right wing resolved in $\mathcal{F}_R$ is given by

$$L_R = \frac{1}{2}\rho C_L(\alpha)\text{sgn}(e_1^TU_R e_3^TU_R)\hat{e}_2U_R\|U_R\|S, \quad (40)$$

Similarly, the drag is given by

$$D_R = -\frac{1}{2}\rho C_D(\alpha)\|U_R\|U_RS. \quad (41)$$

In short, for the special case of small body velocities, the lift and the drag vector of the right wing resolved in $\mathcal{F}_R$ are given by (38)–(39).

Define $r_{cp} \in \mathbb{R}$ such that the above moment is equivalent to the lift and the drag acting on $r_{cp}e_2$, i.e.,

$$r_{cp}e_2 \times \int (dL_R + dD_R) = \int re_2 \times (dL_R + dD_R).$$

For the special case discussed above, one can show

$$r_{cp} = \frac{\int_0^l r^3 c(r)dr}{\int_0^l r^2 c(r)dr} = \frac{S_3}{S_2} = \frac{Sl^3\tilde{r}_3^3}{Sl^2\tilde{r}_2^2} = \frac{\tilde{r}_3^3}{\tilde{r}_2^2}l. \quad (42)$$

Therefore, the span-wise location of the center of pressure of the right wing is given by $r_{cp}e_2$, and the resulting moment about the wing root is

$$M_R = r_{cp}e_2 \times (L_R + D_R), \quad (43)$$

which is resolved in $\mathcal{F}_R$.

Left Wing For the left wing, define the velocity at the chord $r = \tilde{r}_2 l$ be

$$U_L = U_L(\tilde{r}_2 l) = -\tilde{r}_2 \Omega_2 \times e_2. \quad (44)$$

The angle of attack is given by

$$\alpha_L = \cos^{-1}\left(\frac{|e_1^T U_L|}{\|U_L\|}\right) = \sin^{-1}\left(\frac{|e_3^T U_L|}{\|U_L\|}\right). \quad (45)$$

The translational force and the moment for the left wing are

$$L_L = \frac{1}{2} \rho C_L(\alpha) \text{sgn}(e_1^T U_L e_3^T U_L) \hat{e}_2 \|U_L\| S, \quad (46)$$

$$D_L = -\frac{1}{2} \rho C_D(\alpha) \|U_L\| S, \quad (47)$$

$$M_L = -r_{cp} e_2 \times (L_L + D_L), \quad (48)$$

which are resolved in $\mathcal{F}_L$. All of these expressions are consistent with the right wing, except the negative sign in (48).

2.5 Rotational Force

The wing's own rotation may cause additional circulation about the chord that generate a normal force. This is referred to as a rotational force [4]. The expression of the rotational force of the right wing is identical to that of the left wing, except the flipped sign in the corresponding moment. As such, in this section, we do not distinguish the right wing from the left wing with the subscript R .

According to [4], the magnitude of the rotational lift for the infinitesimal wing segment is given by

$$\rho \|U(r)\| C_{\text{rot}} |\dot{\alpha}(r)| c^2(r) dr, \quad (49)$$

where the last part $C_{\text{rot}} |\dot{\alpha}(R)| c^2$ corresponds to the rotational circulation. From (39), the time-derivative of the angle of attack multiplied by $\|U\|$ is given by

$$\begin{aligned} \|U\| \dot{\alpha} &= \frac{1}{\sin \alpha} \left\{ \cos \alpha \frac{U^T \dot{U}}{\|U\|} - \text{sgn}(e_1^T U) (e_1^T \dot{U}) \right\} \\ &= \frac{1}{\cos \alpha} \left\{ -\sin \alpha \frac{U^T \dot{U}}{\|U\|} + \text{sgn}(e_3^T U) (e_3^T \dot{U}) \right\}. \end{aligned} \quad (50)$$

To avoid singularity caused by α , the first expression is used when α is close to $\frac{\pi}{2}$, and the second expression is used when α is close to 0. Also, to mitigate the singularity caused by $\|U\| \ll 1$, we evaluate $\|U\| \dot{\alpha}$ in the numerical implementation, instead of computing $\dot{\alpha}$ directly.

Next, we find the direction of the rotational lift. It is shown that the rotational lift is always perpendicular to the chord line, and consequently it is $\pm e_3$ in $\mathcal{F}_R$. Consider the four cases of $\alpha > 0$ illustrated in Figure 8. For the first two cases of $e_3^T U > 0$, $\dot{\alpha} > 0$ generates the rotational lift along $-e_3$, and for the last two cases of $e_3^T U < 0$, $\dot{\alpha} > 0$ generates the rotational lift along e_3 . Therefore, when $\alpha > 0$, the direction of the rotational lift is $-\text{sgn}(e_3^T U) \text{sgn}(\dot{\alpha}) e_3$.

For the case of $\alpha = 0$, the direction of the rotational lift is illustrated in Figure 9, where the rotational lift is always opposite to $e_3^T \dot{U}$. Therefore, its direction is given by $-\text{sgn}(e_3^T \dot{U}) e_3$.

These two cases for the direction can be combined into

$$\text{sig}_{\text{rot}}(r) e_3, \quad (51)$$

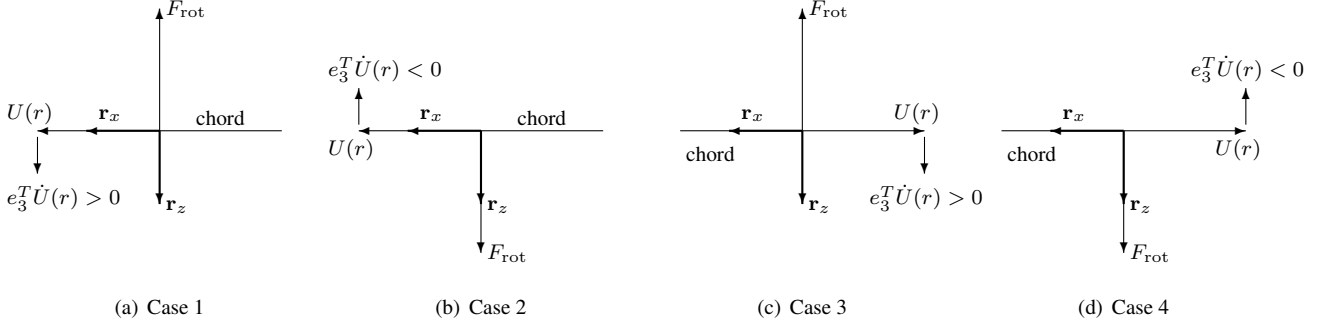


Figure 9: Direction of the rotational lift when $\alpha = 0$

where

$$\text{sig}_{\text{rot}} = \text{sgn}(\alpha(r))\{-\text{sgn}(e_3^T U(r))\text{sgn}(\dot{\alpha}(r))\} + (1 - \text{sgn}(\alpha(r)))\{-\text{sgn}(e_3^T \dot{U}(r))\}, \quad (52)$$

which takes the value of ± 1 . Thus, the rotational lift vector is the product of the above direction and the magnitude (49).

$$dF_{\text{rot}} = \rho \|U(r)\| C_{\text{rot}} |\dot{\alpha}(r)| \text{sgn}_{\text{rot}}(r) e_3 c^2(r) dr, \quad (53)$$

which is integrated to obtain the rotational force

$$F_{\text{rot}} = \int_0^l dF_{\text{rot}}. \quad (54)$$

For the right wing, the above rotational force generate the moment about the wing root as follows.

$$M_{\text{rot}_R} = \int_0^l r e_2 \times dF_{\text{rot}},$$

which is resolved in $\mathcal{F}_R$. For the left wing, the chord is located at $-r e_2$ in $\mathcal{F}_L$, and consequently

$$M_{\text{rot}_L} = - \int_0^l r e_2 \times dF_{\text{rot}}.$$

2.6 Rotational Force for Small Body Velocities

Right Wing Under the assumption of small body velocities, we present an expression for the rotational force that does not require a span-wise integration. As discussed above, when $\|\dot{x}\|, \|\Omega\|, \|v_{\text{wind}}\| \ll 1$, $U_R(r)$ reduces to

$$U_R(r) = r \Omega_R \times e_2,$$

so that $\alpha, \dot{\alpha}$ remains fixed span-wise. Thus,

$$F_{\text{rot}_R} = \rho C_{\text{rot}} \|\Omega_R \times e_2\| |\dot{\alpha}(t)| \text{sgn}_{\text{rot}}(t) e_3 \int_0^l r c^2(r) dr,$$

Substituting (21),

$$F_{\text{rot}_R} = \rho C_{\text{rot}} \|\Omega_R \times e_2\| |\dot{\alpha}(t)| \text{sgn}_{\text{rot}}(t) e_3 l^2 \tilde{c}^2 \tilde{v} \tilde{r}_1(v).$$

Substituting (38), we obtain the rotational force in terms of U_R as

$$F_{\text{rot}_R} = \rho C_{\text{rot}} \|U_R\| |\dot{\alpha}(t)| \text{sgn}_{\text{rot}}(t) e_3 l \bar{c}^2 \frac{\tilde{v} \tilde{r}_1(v)}{\tilde{r}_2}. \quad (55)$$

Define $r_{\text{rot}} \in \mathbb{R}$ such that the moment about the wing root generated by the rotational force is written as

$$M_{\text{rot}_R} = r_{\text{rot}} e_2 \times F_{\text{rot}_R}. \quad (56)$$

Using (21) and (22), one can show

$$r_{\text{rot}} = \frac{\int_0^l r^2 c^2(r) dr}{\int_0^l r c^2(r) dr} = \frac{l \tilde{r}_2^2(v)}{\tilde{r}_1(v)}. \quad (57)$$

Left Wing For the left wing,

$$F_{\text{rot}_L} = \rho C_{\text{rot}} \|U_L\| |\dot{\alpha}(t)| \text{sgn}_{\text{rot}}(t) e_3 l \bar{c}^2 \frac{\tilde{v} \tilde{r}_1(v)}{\tilde{r}_2}, \quad (58)$$

$$M_{\text{rot}_L} = -r_{\text{rot}} e_2 \times F_{\text{rot}_L}. \quad (59)$$

2.7 Numerical Example

We evaluate the aerodynamic forces with the following lift/drag coefficient [4]

$$\begin{aligned} C_L &= 0.225 + 1.58 \sin((2.13\alpha^\circ - 7.2) \frac{\pi}{180}), \\ C_D &= 1.92 - 1.55 \cos((2.04\alpha^\circ - 9.82) \frac{\pi}{180}), \\ C_{\text{rot}} &= \pi(0.75 - \hat{x}_0), \end{aligned}$$

where $\alpha^\circ = \alpha \frac{180}{\pi}$ is the angle of attack in degrees and $\hat{x}_0$ is chosen as $\hat{x}_0 = 0.5$.

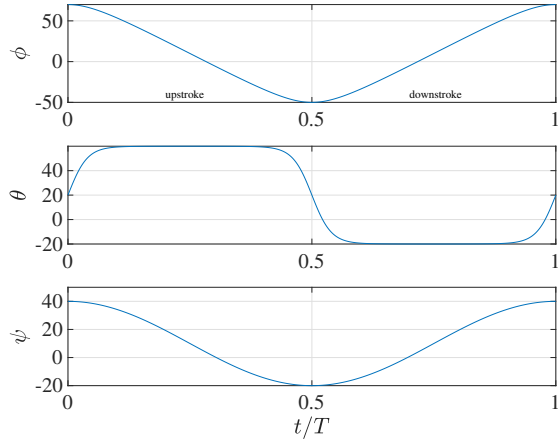
For both of the left wing and the right wing, parameters for the wing kinematics are chosen as

$$\begin{aligned} f &= 10 \text{ Hz}, \quad \beta = 0, \quad \phi_M = 60^\circ, \quad \phi_K = 0.9, \quad \phi_0 = 10^\circ, \\ \theta_M &= 30^\circ, \quad \theta_C = 4, \quad \theta_0 = 10^\circ, \\ \psi_M &= 30^\circ, \quad \psi_N = 1, \quad \psi_a = 0, \quad \psi_0 = 10^\circ. \end{aligned}$$

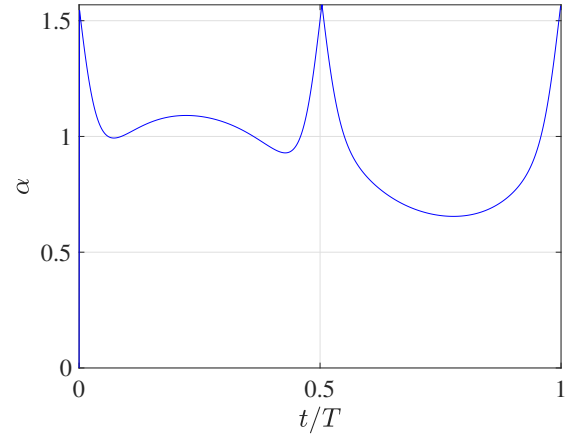
Therefore, the motion of the left wing is symmetric to the right wing. Three cases are considered for $\phi_a = 0$ (symmetric rotation), $\phi_a = 0.3$ (advanced rotation), and $\phi_a = -0.3$ (delayed rotation).

The resulting aerodynamic forces are illustrated in Figures 10–12, where the subfigures (e) show the aerodynamic forces resolved in the wing-fixed frame. It is shown that the first component of the resultant force is almost zero. Thus, the resultant force is normal to the chord.

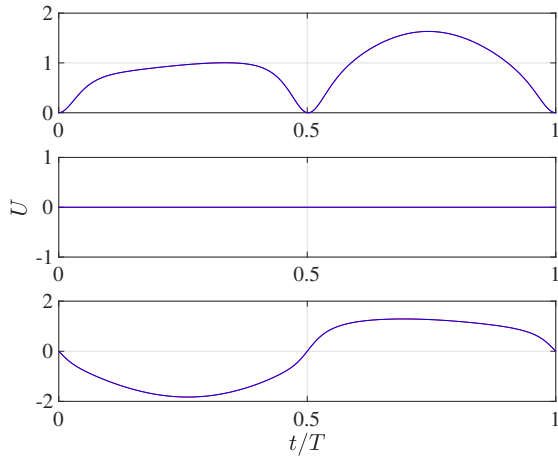
Next, the last subfigures, namely (f) present the aerodynamic forces resolved in $\mathcal{F}_B$. Since $\beta = 0$, the first component is the resultant lift normal to the stroke plane.



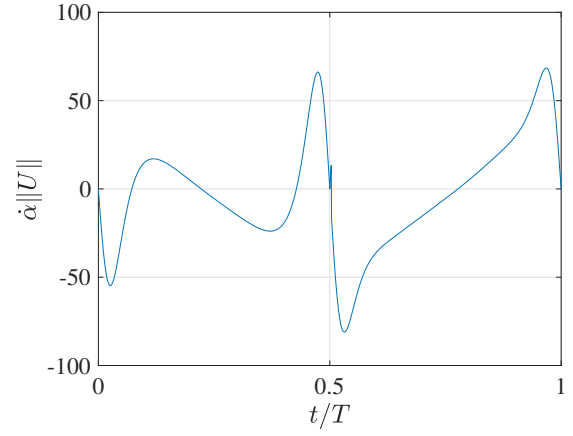
(a) ϕ, θ, ψ



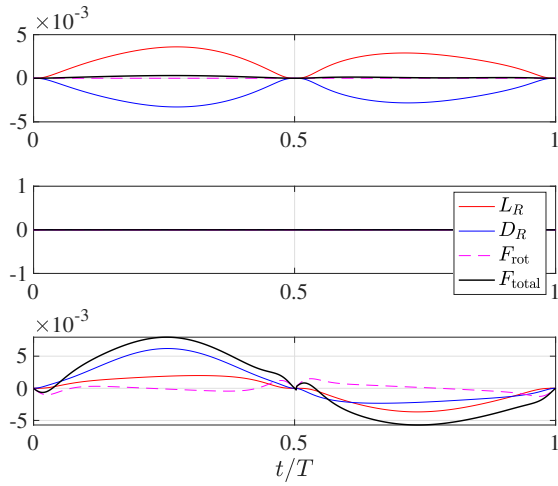
(b) α



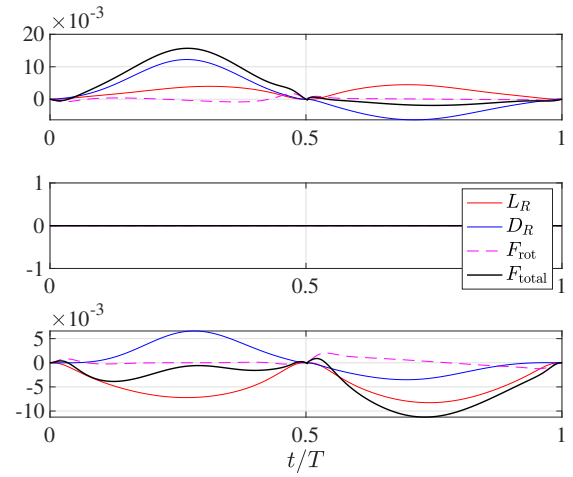
(c) U



(d) $\dot{\alpha}||U||$

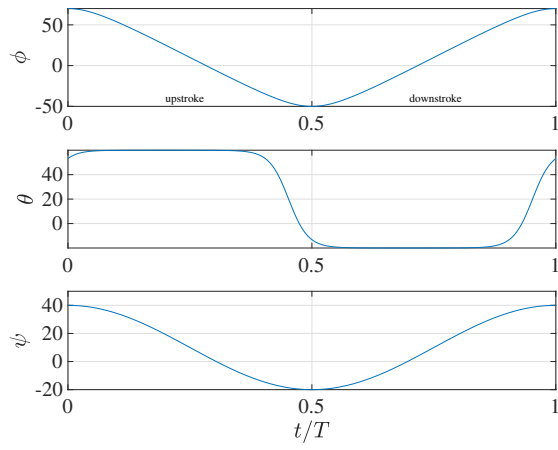


(e) Aerodynamic forces in the wing-fixed frame

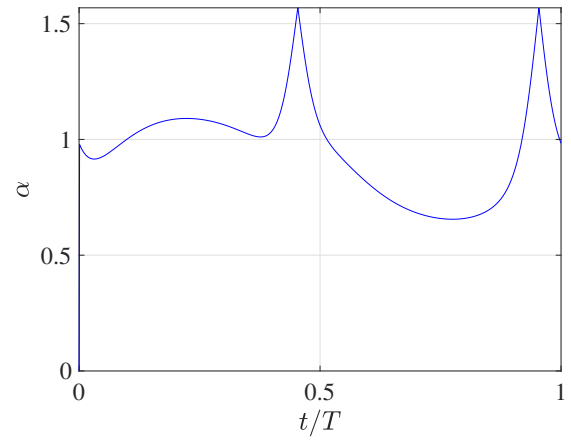


(f) Aerodynamic forces in the body-fixed frame

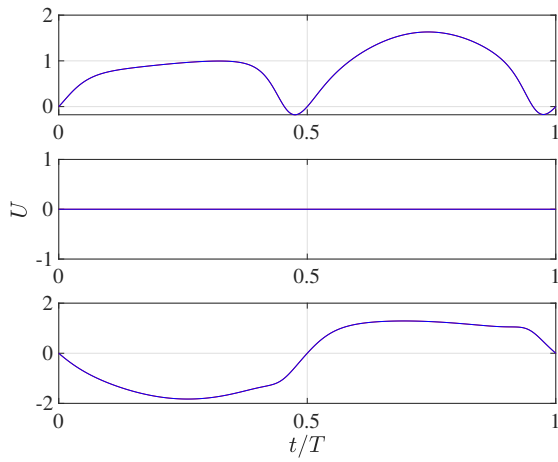
Figure 10: Wing Kinematics in [1]: Symmetric rotation



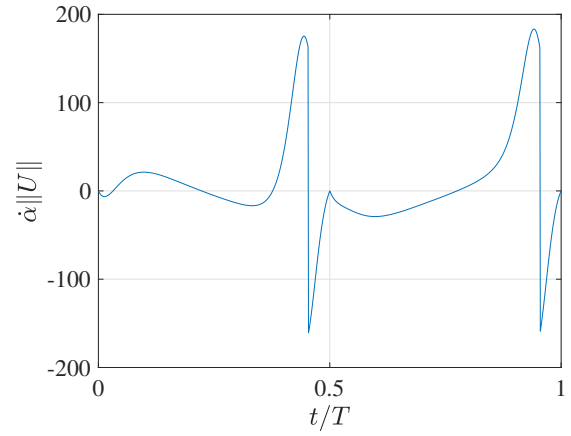
(a) ϕ, θ, ψ



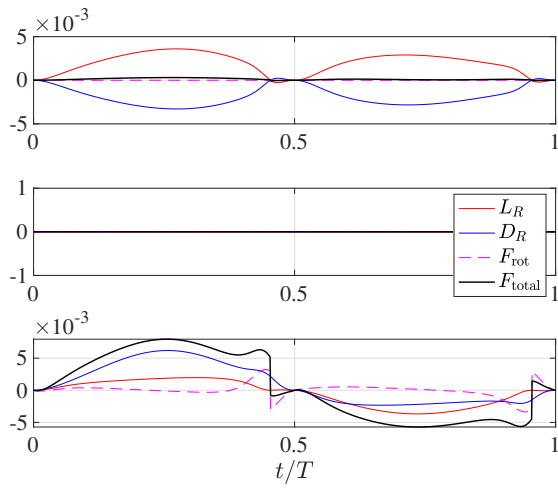
(b) α



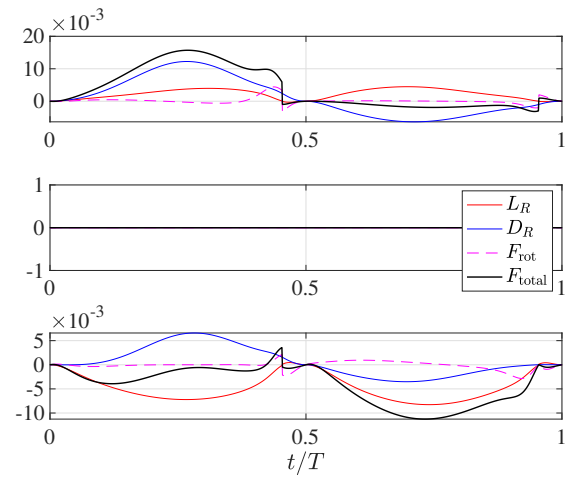
(c) U



(d) $\dot{\alpha}/||U||$

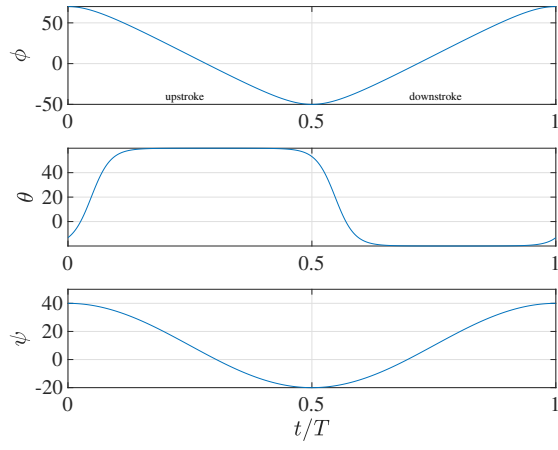


(e) Aerodynamic forces in the wing-fixed frame

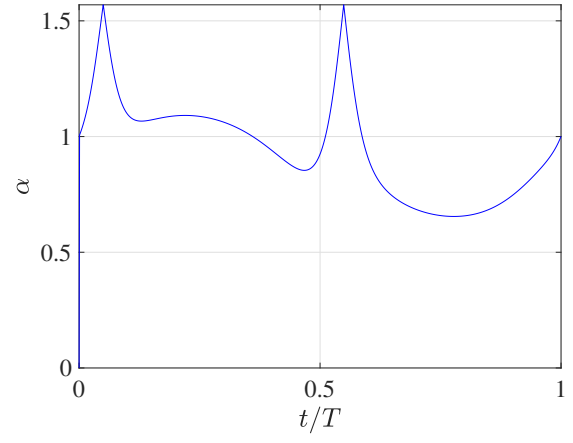


(f) Aerodynamic forces in the body-fixed frame

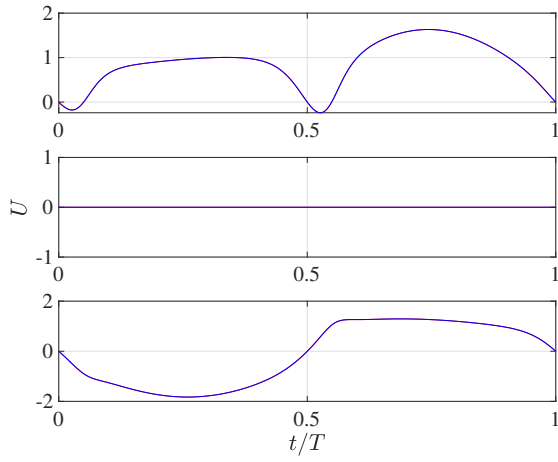
Figure 11: Wing Kinematics in [1]: Advanced rotation



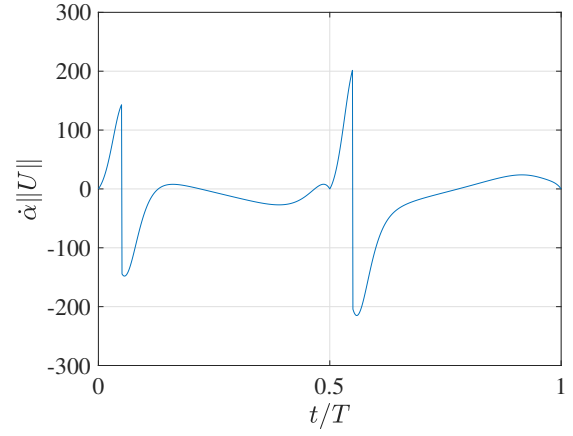
(a) ϕ, θ, ψ



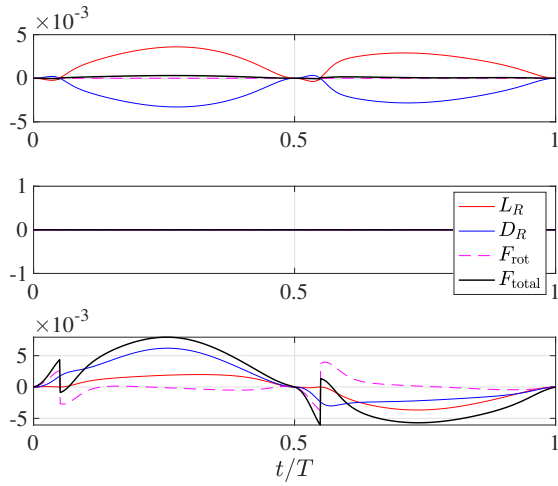
(b) α



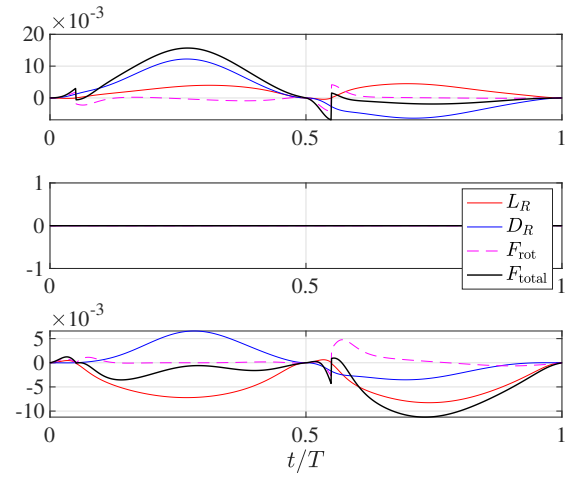
(c) U



(d) $\dot{\alpha}||U||$



(e) Aerodynamic forces in the wing-fixed frame



(f) Aerodynamic forces in the body-fixed frame

Figure 12: Wing Kinematics in [1]: Delayed rotation

3. Flapping Wing UAV Dynamics

3.1 Lagrangian Mechanics on a Lie Group

Consider a dynamic system evolving on an abstract Lie group G . We develop Euler–Lagrange equations for the arbitrary Lie group G , which are utilized later for the flapping wing UAV. The kinematics equation on G is given by

$$\dot{g} = g\xi, \quad (60)$$

for $\xi \in \mathfrak{g}$ corresponding to the left-trivialized velocity. Consequently the tangent bundle TG is identified with $G \times \mathfrak{g}$. Let $\mathbf{J} : G \times \mathfrak{g} \rightarrow \mathfrak{g}^*$ be a symmetric, positive-definite inertia tensor, i.e.,

$$\begin{aligned} \langle \mathbf{J}_g(\xi), \xi \rangle &\geq 0, \\ \langle \mathbf{J}_g(\xi), \xi \rangle &= 0 \Leftrightarrow \xi = 0, \\ \langle \mathbf{J}_g(\xi_1), \xi_2 \rangle &= \langle \mathbf{J}_g(\xi_2), \xi_1 \rangle, \end{aligned}$$

for any $g \in G$ and $\xi, \xi_1, \xi_2 \in \mathfrak{g}$. Also, define $(\mathbf{K}_g(\xi))(\cdot) : G \times \mathfrak{g} \rightarrow \mathfrak{g}^*$ such that

$$\mathbf{T}_e^* \mathbf{L}_g \cdot \mathbf{D}_g \mathbf{J}_g(\xi) \cdot \chi = (\mathbf{K}_g(\xi))(\chi) = \mathbf{K}_g(\xi)\chi. \quad (61)$$

It is straightforward to $\mathbf{K}_g(\xi)$ is a linear operator. Therefore, by selecting a basis of $\mathfrak{g}$, $\mathbf{K}_g(\xi)$ can be represented by a matrix. Intuitively, it is the left-trivialize derivative of $\mathbf{J}_g(\xi)$ with respect to g .

Suppose that the Lagrangian $L : G \times \mathfrak{g} \rightarrow \mathbb{R}$ is given by

$$L(g, \xi) = \frac{1}{2} \langle \mathbf{J}_g(\xi), \xi \rangle - U(g),$$

for a configuration dependent potential $U : SO(3) \rightarrow \mathbb{R}$. We have

$$\begin{aligned} \mathbf{D}_\xi L(g, \xi) &= \mathbf{J}_g(\xi), \\ \frac{d}{dt} \mathbf{D}_\xi L(g, \xi) &= \mathbf{J}_g(\dot{\xi}) + \mathbf{K}_g(\xi)\xi, \\ \mathbf{T}_e^* \mathbf{L}_g \cdot \mathbf{D}_g L(g, \xi) \cdot \chi &= \frac{1}{2} \langle \mathbf{K}_g(\xi)\chi, \xi \rangle - \mathbf{T}_e^* \mathbf{L}_g U(g) \cdot \chi \\ &= \left\{ \frac{1}{2} \mathbf{K}_g^*(\xi)\xi - \mathbf{T}_e^* \mathbf{L}_g U(g) \right\} \cdot \chi, \end{aligned}$$

where $\mathfrak{g}^*$ is identified with $\mathfrak{g}$ with the pairing. Thus, the Euler–Lagrange equations are given by

$$\mathbf{J}_g(\dot{\xi}) + \mathbf{K}_g(\xi)\xi - \text{ad}_\xi^* \cdot \mathbf{J}_g(\xi) - \frac{1}{2} \mathbf{K}_g^*(\xi)\xi + \mathbf{T}_e^* \mathbf{L}_g U(g) = 0. \quad (62)$$

The total energy is

$$E = \frac{1}{2} \langle \mathbf{J}_g(\xi), \xi \rangle + U(g).$$

Its time-derivative is

$$\begin{aligned} \dot{E} &= \langle \dot{\mathbf{J}}_g(\xi), \xi \rangle + \dot{U}(g) \\ &= \frac{1}{2} \langle \dot{\mathbf{J}}_g(\xi), \xi \rangle + \frac{1}{2} \langle \mathbf{J}_g(\xi), \dot{\xi} \rangle + \mathbf{T}_e^* \mathbf{L}_g U(g) \cdot \xi \\ &= \frac{1}{2} \langle \mathbf{J}_g(\dot{\xi}), \xi \rangle + \frac{1}{2} \langle \mathbf{K}_g(\xi)\xi, \xi \rangle + \frac{1}{2} \langle \mathbf{J}_g(\xi), \dot{\xi} \rangle + \mathbf{T}_e^* \mathbf{L}_g U(g) \cdot \xi \\ &= \langle -\mathbf{K}_g(\xi)\xi + \text{ad}_\xi^* \cdot \mathbf{J}_g(\xi) + \frac{1}{2} \mathbf{K}_g^*(\xi)\xi, \xi \rangle + \frac{1}{2} \langle \mathbf{K}_g(\xi)\xi, \xi \rangle \\ &= \langle -\mathbf{K}_g(\xi)\xi, \xi \rangle + \langle \mathbf{J}_g(\xi), \text{ad}_\xi \cdot \xi \rangle + \frac{1}{2} \langle \xi, \mathbf{K}_g(\xi)\xi \rangle + \frac{1}{2} \langle \mathbf{K}_g(\xi)\xi, \xi \rangle = 0, \end{aligned}$$

which shows the energy conservation in the absence of external force or moment.

3.2 Lagrangian Mechanics of Flapping Wing UAV

The configuration of the given flapping wing UAV model is described by $g = (x, R, Q_R, Q_L, Q_A)$. Consequently, the configuration space is a Lie group $G = \mathbb{R}^3 \times \text{SO}(3)^4$. As it is a product of $\mathbb{R}^3$ and four copies of $\text{SO}(3)$, the Lie algebra is simply $\mathfrak{g} = \mathbb{R}^3 \times \mathfrak{so}(3)^4 \simeq \mathbb{R}^3 \times (\mathbb{R}^3)^4$. The kinematics equation is identical to (60) with $\xi = (\dot{x}, \Omega, \Omega_R, \Omega_L, \Omega_A) \in \mathfrak{g}$. We derive the Euler–Lagrange equations for the flapping wing UAV using the preceding results.

Kinetic Energy The kinetic energy of the body is given by

$$T_B = \frac{1}{2}m_B\|\dot{x}\|^2 + \frac{1}{2}\Omega^T J_B \Omega. \quad (63)$$

where $m_B \in \mathbb{R}$ is the mass of the body, and $J_B \in \mathbb{R}^{3 \times 3}$ is the inertia matrix of the body about $\mathcal{F}_B$.

Next, we find the expression for the kinetic energy of the wings and the abdomen. The contribution of the left wing is identical to the right wing and the abdomen, as all of them are essentially a rigid body connected to the thorax via a spherical joint. Therefore, we use the subscript $i \in \{R, L, A\}$ to denote the variables related to a particular wing or the abdomen, which is referred to as $\mathcal{B}_i$.

Consider a mass element dm in $\mathcal{B}_i$, whose location is given by $\nu \in \mathbb{R}^3$ in $\mathcal{F}_i$, i.e., ν is the vector from the joint to the mass element resolved in $\mathcal{F}_i$. Thus, its location from the origin of the inertial frame, resolved in the inertial frame $\mathcal{F}_I$ is $x + R\mu_i + RQ_i\nu$, and its velocity is

$$\dot{x} + R\hat{\Omega}(\mu_i + Q_i\nu) + RQ_i\hat{\Omega}_i\nu.$$

Therefore, the kinetic energy is

$$T_i = \frac{1}{2} \int_{\mathcal{B}_i} \frac{1}{2} \|\dot{x} + R\hat{\Omega}(\mu_i + Q_i\nu) + RQ_i\hat{\Omega}_i\nu\|^2 dm.$$

Let $m_i \in \mathbb{R}$ be the mass of $\mathcal{B}_i$, and let

$$\nu_i = \frac{1}{m_i} \int_{\mathcal{B}_i} \nu dm, \quad (64)$$

$$J_i = \int_{\mathcal{B}_i} \hat{\nu}^T \hat{\nu} dm. \quad (65)$$

Thus $\nu_i \in \mathbb{R}^3$ is the vector from the origin of $\mathcal{F}_i$ to the mass center of $\mathcal{B}_i$ resolved in $\mathcal{F}_i$, and $J_i \in \mathbb{R}^{3 \times 3}$ is the inertia matrix of $\mathcal{B}_i$ about the origin of $\mathcal{F}_i$ resolved in $\mathcal{F}_i$. After rearranging, one can show

$$T_i = \begin{bmatrix} \dot{x} \\ \Omega \\ \Omega_i \end{bmatrix}^T \mathbf{J}_i(R, Q_i) \begin{bmatrix} \dot{x} \\ \Omega \\ \Omega_i \end{bmatrix}.$$

where the configuration-dependent inertia for $\mathcal{B}_i$, namely $\mathbf{J}_i(R, Q_i) \in \mathbb{R}^{9 \times 9}$ is

$$\mathbf{J}_i(R, Q_i) = \begin{bmatrix} m_i I_{3 \times 3} & -m_i R(\hat{\mu}_i + \widehat{Q_i \nu_i}) & -m_i RQ_i \hat{\nu}_i \\ m_i(\hat{\mu}_i + \widehat{Q_i \nu_i})R^T & m_i \hat{\mu}_i^T \hat{\mu}_i + Q_i J_i Q_i^T + m_i(\hat{\mu}_i^T \widehat{Q_i \nu_i} + \widehat{Q_i \nu_i}^T \hat{\mu}_i) & Q_i J_i + m_i \hat{\mu}_i^T Q_i \hat{\nu}_i \\ m_i \hat{\nu}_i Q_i^T R^T & J_i Q_i^T + m_i \hat{\nu}_i^T Q_i^T \hat{\mu}_i & J_i \end{bmatrix}. \quad (66)$$

From (61), the first three element of $(\mathbf{K}_i(\xi))(\chi)$ are given by

$$\begin{aligned} [(\mathbf{K}_i(\xi))(\chi)]_{1:3} &= -m_i R \hat{\eta}(\hat{\mu}_i + \widehat{Q_i \nu_i}) \Omega - m_i R(\widehat{Q_i \hat{\eta}_i \nu_i}) \Omega - m_i R \hat{\eta} Q_i \hat{\nu}_i \Omega_i - m_i R Q_i \hat{\eta}_i \hat{\nu}_i \Omega_i \\ &= 0 \delta x + [m_i R((\hat{\mu}_i + \widehat{Q_i \nu_i}) \Omega)^\wedge + m_i R(Q_i \hat{\nu}_i \Omega_i)^\wedge] \eta \\ &\quad + [-m_i R \hat{\Omega} Q_i \hat{\nu}_i + m_i R Q_i \widehat{\hat{\nu}_i \Omega_i}] \eta_i. \end{aligned}$$

Similarly, repeat this for the remaining rows to construct $\mathbf{K}_i \in \mathbb{R}^{9 \times 9}$,

$$\mathbf{K}_i(R, Q_i, \Omega, \Omega_i) = \begin{bmatrix} 0 & m_i R((\hat{\mu}_i + \widehat{Q_i \nu_i})\Omega + Q_i \hat{\nu}_i \Omega_i)^\wedge & m_i R(-\hat{\Omega} Q_i \hat{\nu}_i + Q_i \hat{\nu}_i \hat{\Omega}_i) \\ 0 & m_i (\hat{\mu}_i + \widehat{Q_i \nu_i}) \widehat{R^T \dot{x}} & m_i \widehat{R^T \dot{x}} Q_i \hat{\nu}_i - Q_i (J_i Q_i^T \Omega)^\wedge + Q_i J_i \widehat{Q_i^T \Omega} \\ & & -m_i \hat{\mu}_i \hat{\Omega} Q_i \hat{\nu}_i - m_i \widehat{\hat{\mu}_i \Omega} Q_i \hat{\nu}_i \\ & & -Q_i \widehat{J_i \Omega_i} + m_i \hat{\mu}_i Q_i \hat{\nu}_i \widehat{\Omega_i} \\ 0 & m_i \hat{\nu}_i Q_i^T \widehat{R^T \dot{x}} & m_i \hat{\nu}_i (Q_i^T R^T \dot{x})^\wedge + J_i \widehat{Q_i^T \Omega} - m_i \hat{\nu} (Q^T \hat{\mu}_i \Omega)^\wedge \end{bmatrix}. \quad (67)$$

The total kinetic energy is the sum of the contributions of the body, the wings and the abdomen, as given by

$$T = T_B + T_R + T_L + T_A.$$

Gravitational Potential Energy The gravitational potential energy of the body is

$$U_B = -m_B g e_3^T x.$$

The gravitational potential energy of $\mathcal{B}_i$ is

$$U_i = -m_i g e_3^T (x + R\mu_i + RQ_i \nu_i).$$

The total potential energy is

$$U = U_B + U_R + U_L + U_A.$$

The negative derivatives of the potential energy, namely $\mathbf{f}_g \in \mathbb{R}^{15}$ corresponds to the gravitational force and moment given by

$$\mathbf{f}_g \triangleq -\mathbf{T}_e^* \mathbf{L}_g U = \begin{bmatrix} (m_B + m_R + m_L + m_A) g e_3 \\ m_R g (\mu_R + Q_R \nu_R)^\wedge R^T e_3 + m_L g (\mu_L + Q_L \nu_L)^\wedge R^T e_3 + m_A g (\mu_A + Q_A \nu_A)^\wedge R^T e_3 \\ m_R g \hat{\nu}_R (Q_R^T R^T e_3) \\ m_L g \hat{\nu}_L (Q_L^T R^T e_3) \\ m_A g \hat{\nu}_A (Q_A^T R^T e_3) \end{bmatrix}, \quad (68)$$

where the total mass is denoted by $m \in \mathbb{R}$,

$$m = m_B + m_R + m_L + m_A. \quad (69)$$

Potential Energy from Torsional Spring Here we formulate a potential energy due to a torsional spring attached to the wing root. Recall $Q_R \in \text{SO}(3)$ transforms the representation of a vector from the right wing frame $\mathcal{F}_R$ to the body-fixed frame $\mathcal{F}_B$. The transform from $\mathcal{F}_R$ to the stroke frame $\mathcal{F}_S$ is given by $\exp(-\beta \hat{e}_2) Q_R$.

The first axis $\mathbf{s}_x$ of $\mathcal{F}_S$ is given by $\mathbf{s}_x = Q_R^T \exp(\beta \hat{e}_2) \mathbf{e}_1$ when resolved in $\mathcal{F}_R$. Since the flapping motion is not supposed to cause any torsion, we project $\mathbf{s}_x$ along $\mathbf{r}_y$ into the plane spanned by $\mathbf{r}_x$ and $\mathbf{r}_z$ to obtain,

$$\mathbf{s}_{x,proj} = (I_{3 \times 3} - e_2 e_2^T) Q_R^T \exp(\beta \hat{e}_2) \mathbf{e}_1.$$

Let γ_R be the angle that is proportional to the torque caused by the spring. It is defined as the angle between $\mathbf{r}_x$ the above projection, i.e.,

$$\cos \gamma_R = \frac{e_1^T \mathbf{s}_{x,proj}}{\|\mathbf{s}_{x,proj}\|} = \frac{e_1^T \mathbf{s}_x}{\|\mathbf{s}_{x,proj}\|}, \quad (70)$$

$$\sin \gamma_R = \frac{e_3^T \mathbf{s}_{x,proj}}{\|\mathbf{s}_{x,proj}\|} = \frac{e_3^T \mathbf{s}_x}{\|\mathbf{s}_{x,proj}\|}. \quad (71)$$

Therefore,

$$\gamma_R(Q_R) = \tan^{-1} \frac{e_3^T s_x}{e_1^T s_x}. \quad (72)$$

For the particular 1-3-2 Euler-angles chosen for Q_R , we have $e_1^T s_x = \cos \psi_R \cos \theta_R$ and $e_3^T s_x = \cos \psi_R \sin \theta_R$. Therefore, the expression of γ_R defined previously is equivalent to the pitch angle, i.e., $\gamma_R = \theta_R$.

The potential energy due to the torsional spring can be written as,

$$U_{t,R}(Q_R) = \frac{1}{2} \kappa_t \theta_R^2(Q_R) \quad (73)$$

for $\kappa_t > 0$. Let $\tau_{t,R} \in \mathbb{R}^3$ be the torque generated by the above potential. It can be obtained by

$$\tau_{t,R} = -\mathbf{T}_e \mathbf{L}_{Q_R} U_{t,R}(Q_R),$$

or equivalently

$$-\tau_{R,t}^T \chi_R = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} U_{R,t}(Q_R(I_{3 \times 3} + \epsilon \hat{\chi}_R)),$$

which is satisfied for any $\chi_R \in \mathbb{R}^3$. This expression yields

$$\begin{aligned} \tau_{R,t}(Q_R) &= \frac{\kappa_t \theta_R(Q_R)}{(1 + \tan^2(\theta_R(Q_R)))(e_1^T s_x)^2} ((e_2^T s_x) s_x - e_2) \\ &= \kappa_t \theta_R(Q_R) \frac{(e_2^T s_x) s_x - e_2}{(e_1^T s_x)^2 + (e_3^T s_x)^2} \\ &= -\kappa_t \frac{\theta_R(Q_R)}{\|\hat{s}_x^2 e_2\|} \frac{\hat{s}_x^2 e_2}{\|\hat{s}_x^2 e_2\|}, \end{aligned} \quad (74)$$

where $-\hat{s}_x^2 e_2$ corresponds to the projection of e_2 onto the plane normal to s_x . As all of these are resolved in $\mathcal{F}_R$, the direction of the torque is the projection of $\mathbf{r}_y$ to the plane spanned by $\mathbf{s}_y$ and $\mathbf{s}_z$. Next, the magnitude of the torque corresponds to $\kappa_t \theta_R(Q_R) / \|\hat{s}_x^2 e_2\|$. Note, it is not proportional to $\theta_R(Q_R)$, but the magnitude of the torque is scaled by the factor

$$\frac{1}{\|\hat{s}_x^2 e_2\|} = \frac{1}{\|s_{x,proj}\|} = \frac{1}{\sqrt{1 - (e_2^T s_x)^2}}. \quad (75)$$

This implies that the torque increases as the projection of $\mathbf{s}_x$ to the $\mathbf{r}_x - \mathbf{r}_z$ plane, or the projection of $\mathbf{r}_y$ to the $\mathbf{s}_y - \mathbf{s}_z$ plane become shorter. The reason is that as those vectors become shorter, the angle θ_R becomes more sensitive to the attitude of the right wing represented by Q_R . In short, for this formulation, the potential energy due to the torsion is proportional to the pitch angle squared. But the magnitude of the torque is not directly proportional to the pitch angle.

Similarly a torsional spring at the left wing with a potential,

$$U_{L,t}(Q_L) = \frac{1}{2} \kappa_t \theta_L^2(Q_L) \quad (76)$$

yields a torque,

$$\tau_{L,t}(Q_L) = \frac{\kappa_t \theta_L(Q_L)}{(1 + \tan^2(\theta_L(Q_L)))(e_1^T s_x)^2} ((e_2^T s_x) s_x - e_2) \quad (77)$$

where $s_x = Q_L^T \exp(\beta \hat{e}_2) e_1$.

Hence, the total potential energy from the previous subsection is updated by these two additional values as $U + U_{t,R} + U_{t,L}$. The generalized force due to the torsional spring is denoted by

$$\mathbf{f}_t = \begin{bmatrix} 0 \\ 0 \\ \tau_{t,R} \\ \tau_{t,L} \\ 0 \end{bmatrix}. \quad (78)$$

Virtual Work Consider an infinitesimal aerodynamic force $dF(\nu) \in \mathbb{R}^3$ acting at the location of $\nu \in \mathbb{R}^3$ of the wing or the abdomen. In the inertial frame, the location of the force is given by $x + R\mu_i + RQ_i\nu$. Thus, the corresponding virtual work due to the aerodynamic force is

$$\int_{\mathcal{B}_i} \delta(x + R\mu_i + RQ_i\nu) \cdot RQ_i dF(\nu) d\nu = (\delta x + R\hat{\eta}\mu_i) \cdot RQ_i \int_{\mathcal{B}_i} dF(\nu) + \eta_i \cdot \int_{\mathcal{B}_i} \nu \times dF(\nu).$$

Let the resultant aerodynamic force and the moment about the wing root or the joint connecting the abdomen be

$$F_i = \int_{\mathcal{B}_i} dF_i(\nu), \quad M_i = \int_{\mathcal{B}_i} \nu \times dF_i(\nu).$$

Also, let $\tau_i \in \mathbb{R}^3$ be the control torque exerted to the wing root or at the joint connecting the abdomen, resolved in the body-fixed frame. As it is an internal torque, there will be a reactive torque, namely $-\tau_i$ exerted to the body.

The total virtual work due to the aerodynamic force and the control torque can be written as

$$\begin{aligned} \delta\mathcal{W} &= \sum_{i \in \{R,L,A\}} \delta x \cdot RQ_i F_i + \eta \cdot (\mu_i \times Q_i F_i - \tau_i) + \eta_i \cdot (M_i + Q_i^T \tau_i) \\ &= (\mathbf{f}_a + \mathbf{f}_\tau) \cdot \xi, \end{aligned} \quad (79)$$

where $\mathbf{f}_a, \mathbf{f}_\tau \in \mathbb{R}^{15}$ denote the contributions of the aerodynamic forces and the control torque, respectively. They are given by

$$\mathbf{f}_a = \begin{bmatrix} RQ_R F_R + RQ_L F_L + RQ_A F_A \\ \hat{\mu}_R Q_R F_R + \hat{\mu}_L Q_L F_L + \hat{\mu}_A Q_A F_A \\ M_R \\ M_L \\ M_A \end{bmatrix}, \quad (80)$$

$$\mathbf{f}_\tau = \begin{bmatrix} 0 \\ -\tau_R - \tau_L - \tau_A \\ Q_R^T \tau_R \\ Q_L^T \tau_L \\ Q_A^T \tau_A \end{bmatrix}. \quad (81)$$

Euler-Lagrange Equations The inertia tensor for the complete UAV, namely $\mathbf{J}_g(\xi) \in \mathbb{R}^{15 \times 15}$ is written as

$$\mathbf{J}_g(\xi) = \begin{bmatrix} m_B I_{3 \times 3} + \mathbf{J}_{R11} + \mathbf{J}_{L11} + \mathbf{J}_{A11} & \mathbf{J}_{R12} + \mathbf{J}_{L12} + \mathbf{J}_{A12} & \mathbf{J}_{R13} & \mathbf{J}_{L13} & \mathbf{J}_{A13} \\ \cdot & J_B + \mathbf{J}_{R22} + \mathbf{J}_{L22} + \mathbf{J}_{A22} & \mathbf{J}_{R23} & \mathbf{J}_{L23} & \mathbf{J}_{A23} \\ \cdot & \cdot & \mathbf{J}_{R33} & 0 & 0 \\ \cdot & \cdot & 0 & \mathbf{J}_{L33} & 0 \\ \cdot & \cdot & 0 & 0 & \mathbf{J}_{A33} \end{bmatrix}, \quad (82)$$

where the subscript ij refers to the 3×3 , i, j -th block of the corresponding matrix, and the unspecified blocks are chosen such that $\mathbf{J}_g$ becomes symmetric. The derivatives of the inertia are expressed by the following matrix $\mathbf{K}_g \in \mathbb{R}^{15 \times 15}$ according to (61),

$$\mathbf{K}_g = \begin{bmatrix} 0 & \mathbf{K}_{R12} + \mathbf{K}_{L12} + \mathbf{K}_{A12} & \mathbf{K}_{R13} & \mathbf{K}_{L13} & \mathbf{K}_{A13} \\ 0 & \mathbf{K}_{R22} + \mathbf{K}_{L22} + \mathbf{K}_{A22} & \mathbf{K}_{R23} & \mathbf{K}_{L23} & \mathbf{K}_{A23} \\ 0 & \mathbf{K}_{R32} & \mathbf{K}_{R33} & 0 & 0 \\ 0 & \mathbf{K}_{L32} & 0 & \mathbf{K}_{L33} & 0 \\ 0 & \mathbf{K}_{A32} & 0 & 0 & \mathbf{K}_{A33} \end{bmatrix}. \quad (83)$$

The co-adjoint operator corresponds to the following matrix,

$$\text{ad}_\nu^* = \text{diag}[0_{3 \times 3}, -\hat{\Omega}, -\hat{\Omega}_R, -\hat{\Omega}_L, -\hat{\Omega}_A]. \quad (84)$$

The Euler–Lagrange equations are given according to (62) as

$$\mathbf{J}_g(\dot{\xi}) - \text{ad}_\xi^* \cdot \mathbf{J}_g(\xi) + \mathbf{L}_g(\xi)\xi = \mathbf{f}_a + \mathbf{f}_g + \mathbf{f}_t + \mathbf{f}_\tau. \quad (85)$$

The matrix $\mathbf{L}_g(\xi) = \mathbf{K}_g(\xi) - \frac{1}{2}\mathbf{K}_g^T(\xi) \in \mathbb{R}^{15 \times 15}$ represents the effects of the dependency of the inertia on the configuration, and it is more explicitly given in the appendix.

The above Euler–Lagrange equations are driven by the control torque acting on the joint of the wings and the joint of the abdomen, and they can be simulated by any ODE solver with the initial condition $(g(0), \xi(0))$ and the trajectory of $(\tau_R(t), \tau_L(t), \tau_A(t))$.

3.3 Prescribed Wing Kinematics and Abdomen Attitude

In case the flapping motion of the wings and the relative motion of the abdomen are prescribed, i.e., $Q_R(t), Q_L(t), Q_A(t)$ are given as a function of time, a reduced set of equations for (x, R) can be constructed as follows.

Let the configuration variables be decomposed into two parts, $g = (g_1, g_2)$ and $\xi = (\xi_1, \xi_2)$ with

$$g_1 = (x, R), \quad \xi_1 = [\dot{x}, \Omega], \quad (86)$$

$$g_2 = (Q_R, Q_L, Q_A), \quad \xi_2 = [\Omega_R, \Omega_L, \Omega_A]. \quad (87)$$

According to the assumption $(g_2, \xi_2, \dot{\xi}_2)$ are already known. We construct differential equations for ξ_1 as follows. The Euler–Lagrange equation (85) can be decomposed accordingly as

$$\mathbf{J}_{11}\dot{\xi}_1 + \mathbf{J}_{12}\dot{\xi}_2 - \text{ad}_{\xi_1}^* \cdot (\mathbf{J}_{11}\xi_1 + \mathbf{J}_{12}\xi_2) + \mathbf{L}_{11}\xi_1 + \mathbf{L}_{12}\xi_2 = \mathbf{f}_{a1} + \mathbf{f}_{g1} + \mathbf{f}_{\tau_1}, \quad (88)$$

$$\mathbf{J}_{21}\dot{\xi}_1 + \mathbf{J}_{22}\dot{\xi}_2 - \text{ad}_{\xi_2}^* \cdot (\mathbf{J}_{21}\xi_1 + \mathbf{J}_{22}\xi_2) + \mathbf{L}_{21}\xi_1 + \mathbf{L}_{22}\xi_2 = \mathbf{f}_{a2} + \mathbf{f}_{g2} + \mathbf{f}_{t2} + \mathbf{f}_{\tau_2}, \quad (89)$$

where the matrices $\mathbf{J}_g, \mathbf{L} \in \mathbb{R}^{15 \times 15}$ are decomposed after the first six rows and columns. For example, $\mathbf{J}_{11} \in \mathbb{R}^{6 \times 6}$ is the first 6×6 diagonal block of $\mathbf{J}_g$, and the remaining parts of the first 6 rows are defined as $\mathbf{J}_{12} \in \mathbb{R}^{6 \times 9}$. We cannot integrate (88) separately, as it is coupled with (89) through the unknown (τ_R, τ_L, τ_A) .

From (81),

$$\mathbf{f}_{\tau_1} = \begin{bmatrix} 0 & 0 & 0 \\ -Q_R & -Q_L & -Q_A \end{bmatrix} \mathbf{f}_{\tau_2} \triangleq C \mathbf{f}_{\tau_2},$$

for $C(Q_R, Q_L, Q_A) \in \mathbb{R}^{6 \times 9}$. Multiply (89) by C , and subtract it from (88) to obtain

$$\begin{aligned} & (\mathbf{J}_{11} - C\mathbf{J}_{21})\dot{\xi}_1 - (\text{ad}_{\xi_1}^* \mathbf{J}_{11} - C\text{ad}_{\xi_2}^* \mathbf{J}_{21})\xi_1 + (\mathbf{L}_{11} - C\mathbf{L}_{21})\xi_1 \\ & = -(\mathbf{J}_{12} - C\mathbf{J}_{22})\dot{\xi}_2 + (\text{ad}_{\xi_1}^* \mathbf{J}_{12} - C\text{ad}_{\xi_2}^* \mathbf{J}_{22})\xi_2 - (\mathbf{L}_{12} - C\mathbf{L}_{22})\xi_2 \\ & \quad + \mathbf{f}_{a1} + \mathbf{f}_{g1} - C(\mathbf{f}_{a2} + \mathbf{f}_{g2} + \mathbf{f}_{t2}). \end{aligned} \quad (90)$$

Now, the unknown (τ_R, τ_L, τ_A) is eliminated, and for given $(g_2, \xi_2, \dot{\xi}_2)$ the above can be simulated to construct (g_1, ξ_1) . Once (g_1, ξ_1) is obtained, it can be substituted into (89) to compute (τ_R, τ_L, τ_A) that the torque at the thorax required to rotate the wings and the abdomen as described.

3.4 Prescribed Flapping/Deviation Angle and Abdomen Attitude

Suppose that a part of the wing kinematics, represented by the flapping angle and the deviation angle, is prescribed. In other words, we have $(\phi_R(t), \psi_R(t), \phi_L(t), \psi_L(t))$ prescribed as a function of time. The pitching angle $(\theta_R(t), \theta_L(t))$ should be determined by the equations of motion (86), (87).

First, we assume that there is a torsional spring in the root of the wing. In the above, we modeled a torsional spring through its potential energy proportional to the pitch angle squared. Here, we consider an alternative formulation that is directly proportional to the pitch angle without considering its potential, as given by

$$\mathbf{f}_t = \begin{bmatrix} 0_{3 \times 1} \\ 0_{3 \times 1} \\ -\kappa_t(\theta_R - \theta_0)e_2 \\ -\kappa_t(\theta_L - \theta_0)e_2 \\ 0_{3 \times 3} \end{bmatrix}, \quad (91)$$

for a torsional spring coefficient $\kappa_t > 0$ and a pitch offset θ_0 . Also, it is assumed that there is no internal, control torque along the pitching axis, $\mathbf{r}_y$ and $\mathbf{l}_y$. Since the internal control torque τ_R is resolved in $\mathcal{F}_B$, it is $Q_R^T \tau_R$ when resolved in $\mathcal{F}_R$. Therefore, the assumption of zero pitching moment yields

$$e_2^T Q_R^T \tau_R = 0. \quad (92)$$

Similarly, for the left wing

$$e_2^T Q_L^T \tau_L = 0. \quad (93)$$

Now, let's consider how the above (91)–(93) can be integrated with the complete equations of motion (86) and (87). The number of equations that we have is 17 (6 from (86), 9 from (87), and 2 from (92)–(93)). Note that Ω_i is determined by $(\theta_i, \psi_i, \dot{\phi}_i, \dot{\theta}_i, \dot{\psi}_i)$ for $i \in \{R, L\}$ through (5) or (8). Since (ϕ_i, ψ_i) are given as a function of time, we can determine Ω_i if $(\theta_i, \dot{\theta}_i)$ are available. Similarly, $\dot{\Omega}_i$ can be determined by $(\theta_i, \dot{\theta}_i, \ddot{\theta}_i)$. Thus, the number of unknown variables in (86), (87), (92), and (93) is also 17: 6 of $\dot{\xi}_1$, 9 of (τ_L, τ_R, τ_A) , and 2 of $(\ddot{\theta}_R, \ddot{\theta}_L)$. Here, $\dot{\theta}_i$ is not counted as one of the unknown variables, as it can be obtained by integrating $\ddot{\theta}_i$ with an initial condition $\dot{\theta}_i(0)$.

Therefore, the equations (86), (87), (92), and (93) can be solved together for $\dot{\xi}_1$, (τ_L, τ_R, τ_A) , and $(\ddot{\theta}_R, \ddot{\theta}_L)$. While the resulting expression would be quite involved due to the complexities of (5) and (8), the unknown variables appear linearly. As such, all of the equations may be arranged in the form of

$$\begin{bmatrix} \dots & \dots \\ \dots & \dots \\ \dots & \dots \\ \dots & \dots \end{bmatrix} \begin{bmatrix} \dot{\xi}_1 \\ \ddot{\theta}_R \\ \ddot{\theta}_L \\ \tau_R \\ \tau_L \\ \tau_A \end{bmatrix} = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix},$$

which can be integrated numerically with the kinematics (60).

To obtain these equations we expand the wing angular accelerations as

$$\dot{\Omega}_i = T_i[\ddot{\phi}_i, \ddot{\theta}_i, \ddot{\psi}_i]^T + (\dot{\theta}_i T_{\theta_i} + \dot{\psi}_i T_{\psi_i})[\dot{\phi}_i, \dot{\theta}_i, \dot{\psi}_i]^T$$

for $i = \{R, L\}$ and

$$T_i = \begin{bmatrix} c\psi_i c\theta_i & 0 & s\theta_i \\ s\psi_i & 1 & 0 \\ c\psi_i s\theta_i & 0 & -c\theta_i \end{bmatrix} \quad T_{\theta_i} = \begin{bmatrix} -c\psi_i s\theta_i & 0 & c\theta_i \\ 0 & 0 & 0 \\ c\psi_i c\theta_i & 0 & s\theta_i \end{bmatrix} \quad T_{\psi_i} = \begin{bmatrix} -c\theta_i s\psi_i & 0 & 0 \\ c\psi_i & 0 & 0 \\ -s\psi_i s\theta_i & 0 & 0 \end{bmatrix}$$

are the corresponding transformation matrices for the right wing (here c is $\cos$ while s is $\sin$). If we multiply the first and last rows of all the three matrices by (-1) , we will have the values for the left wing (i.e., $T_L = \text{diag}[-1, 1, -1] T_R$ and so on).

Denoting $\Upsilon_i = [\phi_i, \theta_i, \psi_i]^T$, we can express, $\dot{\Omega}_R - T_R[:, 2]\ddot{\theta}_R = T_R[:, 1]\ddot{\phi}_R + T_R[:, 3]\ddot{\psi}_R + (\dot{\theta}_R T_{\theta_R} + \dot{\psi}_R T_{\psi_R})\dot{\Upsilon}_R$. So the known value of $\dot{\xi}_2 = [\dot{\Omega}_R, \dot{\Omega}_L, \dot{\Omega}_A]^T$ is

$$\dot{\xi}_{2,\phi,\psi} = \begin{bmatrix} T_R[:, 1]\ddot{\phi}_R + T_R[:, 3]\ddot{\psi}_R + (\dot{\theta}_R T_{\theta_R} + \dot{\psi}_R T_{\psi_R})\dot{\Upsilon}_R \\ T_L[:, 1]\ddot{\phi}_L + T_L[:, 3]\ddot{\psi}_L + (\dot{\theta}_L T_{\theta_L} + \dot{\psi}_L T_{\psi_L})\dot{\Upsilon}_L \\ \dot{\Omega}_A \end{bmatrix} \quad \text{with} \quad \dot{\xi}_2 = \begin{bmatrix} T_R[:, 2] & 0_{3 \times 1} \\ 0_{3 \times 1} & T_L[:, 2] \\ 0_{3 \times 1} & 0_{3 \times 1} \end{bmatrix} \begin{bmatrix} \ddot{\theta}_R \\ \ddot{\theta}_L \end{bmatrix} + \dot{\xi}_{2,\phi,\psi}$$

Denoting the above coefficient matrix by $T_{\tilde{\theta}}$, we simplify the 17 equations to obtain the following,

$$\begin{bmatrix} J_{11} & J_{12}T_{\tilde{\theta}} & \begin{bmatrix} 0_{3 \times 9} \\ I_{3 \times 3} & I_{3 \times 3} & I_{3 \times 3} \end{bmatrix} \\ J_{21} & J_{22}T_{\tilde{\theta}} & -\text{diag}[Q_R^T, Q_L^T, Q_A^T] \\ 0_{2 \times 8} & \begin{bmatrix} e_2^T Q_R^T & 0_{1 \times 3} \\ 0_{1 \times 3} & e_2^T Q_L^T \end{bmatrix} & 0_{2 \times 3} \end{bmatrix} \begin{bmatrix} \dot{\xi}_1 \\ \ddot{\theta}_R \\ \ddot{\theta}_L \\ \tau_R \\ \tau_L \\ \tau_A \end{bmatrix} = \begin{bmatrix} \text{ad}_{\xi_1}^* \cdot (\mathbf{J}_{11}\xi_1 + \mathbf{J}_{12}\xi_2) - \mathbf{L}_{11}\xi_1 - \mathbf{L}_{12}\xi_2 + \mathbf{f}_{a_1} + \mathbf{f}_{g_1} - J_{12}\dot{\xi}_{2,\phi,\psi} \\ \text{ad}_{\xi_2}^* \cdot (\mathbf{J}_{21}\xi_1 + \mathbf{J}_{22}\xi_2) - \mathbf{L}_{21}\xi_1 - \mathbf{L}_{22}\xi_2 + \mathbf{f}_{a_2} + \mathbf{f}_{g_2} + \mathbf{f}_{t_2} - J_{22}\dot{\xi}_{2,\phi,\psi} \\ 0_{2 \times 1} \end{bmatrix}$$

Next, we can eliminate τ s from (88) and (89) as illustrated in (90). Furthermore, the equations corresponding to zero pitching moment imply that the 2nd and 5th component of $\mathbf{f}_{\tau_2}$ are 0. So we can also express these constraints by eliminating τ s from (89) by pre-multiplying the whole equation with $E_{R,L} = [e_{2,9}, e_{5,9}]^T$. Here $e_{2,9}, e_{5,9}$ are the 9-dimensional unit vectors, implying $E_{R,L}\mathbf{f}_{\tau_2} = 0_{2 \times 1}$. Hence we obtain 8 independent equations as,

$$\begin{bmatrix} J_{11} - CJ_{21} & (J_{12} - CJ_{22})T_{\tilde{\theta}} \\ E_{R,L}J_{21} & E_{R,L}J_{22}T_{\tilde{\theta}} \end{bmatrix} \begin{bmatrix} \dot{\xi}_1 \\ \ddot{\theta}_R \\ \ddot{\theta}_L \end{bmatrix} = \begin{bmatrix} ((\text{ad}_{\xi_1}^* \mathbf{J}_{11} - C\text{ad}_{\xi_2}^* \mathbf{J}_{21})\xi_1 - (\mathbf{L}_{11} - C\mathbf{L}_{21})\xi_1 \\ + (\text{ad}_{\xi_1}^* \mathbf{J}_{12} - C\text{ad}_{\xi_2}^* \mathbf{J}_{22})\xi_2 - (\mathbf{L}_{12} - C\mathbf{L}_{22})\xi_2 \\ + \mathbf{f}_{a_1} + \mathbf{f}_{g_1} - C(\mathbf{f}_{a_2} + \mathbf{f}_{g_2} + \mathbf{f}_{t_2}) - (J_{12} - CJ_{22})\dot{\xi}_{2,\phi,\psi}) \\ E_{R,L}(\text{ad}_{\xi_2}^* \cdot (\mathbf{J}_{21}\xi_1 + \mathbf{J}_{22}\xi_2) - \mathbf{L}_{21}\xi_1 - \mathbf{L}_{22}\xi_2 + \mathbf{f}_{a_2} + \mathbf{f}_{g_2} + \mathbf{f}_{t_2} - J_{22}\dot{\xi}_{2,\phi,\psi}) \end{bmatrix}$$

Implementation Details Once we solve this relation, we can substitute back to compute (τ_R, τ_L, τ_A) from (89) as implemented in modules / eom_QS_xR_pitch.m. This process of trajectory integration is further sped up using MATLAB coder by initially running modules / mex_crgr_xR_ab / generate_mex to obtain the required mex-file. For instance, the setup for obtaining a periodic hover trajectory is described below (in sim_QS_xR_pitch_hover.m) which is quite similar to the setup in ‘effects of abdomen motion’ paper.

The parameters being optimized over characterize FWUAV flapping and deviation angles and abdomen undulation along with the initial conditions. More specifically, this is formulated as follows.

- The objective function is

$$J = w_1 \int_0^T |E(t)|dt + w_2 \int_0^T |\dot{E}(t)|dt, \quad (94)$$

where $w_1, w_2 \in \mathbb{R}$, and $E(t) = \frac{1}{2}m \|\dot{x}(t)\|^2 - mge_3^T x(t)$ is the sum of body kinetic energy and the gravitational potential energy.

- The states evolving freely consist of $X(t) = \{x(t), R(t), \dot{x}(t), \Omega(t), \theta_R(t), \theta_L(t), \dot{\theta}_R(t), \dot{\theta}_L(t)\}$. We ensure periodic motion by imposing the constraints: $X(0) = X(T)$. It is also assumed that the motion of wings is symmetric to each other in this simple maneuver.
- The optimization parameters are given by
 - flapping frequency: f and stroke plane angle: β
 - wing kinematics: $(\phi_m, \phi_K, \phi_0), (\psi_m, \psi_0, \psi_a)$
 - abdomen undulation : $(\theta_{A_m}, \theta_{A_0}, \theta_{A_a})$
 - initial translational velocity: $\dot{x}(0) \in \mathbb{R}^3$
 - initial attitude, angular velocity along 2nd axis: $\theta_B(0)$ s.t. $R(0) = \exp(\theta_B(0)\hat{e}_2)$, $\Omega_2(0) = \langle \Omega, e_2 \rangle$; taken to be 0 along other axes for this simple maneuver
 - initial pitch, pitch rate and pitch offset: $\theta(0), \dot{\theta}(0), \theta_0$

Meanwhile, the torsional spring coefficient, κ_t , is approximately estimated from the physical characteristics of previous optimization results.

The results that we obtained are in `sim_data / sim_QS_xR_pitch_hover.mat` and the corresponding trajectories are shown in Figure 13.

3.5 Prescribed Wing Kinematics and Abdomen/Body Attitude

Next, we consider the case where the attitude of the thorax is prescribed additionally. As such all of $(R(t), Q_R(t), Q_L(t), Q_A(t))$ are given as a function of time. We construct a equation of motion for the position x .

The first block of (85) is written as

$$m\ddot{x} + \sum_{i \in \{R, L, A\}} \{ \mathbf{J}_{i_{12}} \dot{\Omega} + \mathbf{J}_{i_{13}} \dot{\Omega}_i + \mathbf{K}_{i_{12}} \Omega + \mathbf{K}_{i_{13}} \Omega_i \} = R \sum_{i \in \{R, L, A\}} Q_i F_i + mge_3, \quad (95)$$

which can be numerically integrated to construct $(x, \dot{x})$. The control torque (τ_R, τ_L, τ_A) can be constructed by (89).

3.6 Linearization of dynamics

Finally, we need to analyze the characteristics of motion of the model. And to understand the stability properties of a state trajectory, we can use the linearized version of the corresponding dynamics.

For the Euler-Lagrange equation in (85), consider variations of the form,

$$\delta g = (\delta x, \delta R, \delta Q_R, \delta Q_L, \delta Q_A) \in \mathbf{T}_g \mathbf{G}, \quad \delta \xi = (\delta \dot{x}, \delta \Omega, \delta \Omega_R, \delta \Omega_L, \delta \Omega_A) \in \mathbf{T}_e \mathbf{g} \simeq \mathbb{R}^{15}$$

in which δg can be identified with $(\delta x, \eta, \eta_R, \eta_L, \eta_A) \in \mathbb{R}^{15} \simeq \mathbf{g}$ since the variation in attitude can be expressed as $\delta R = R\hat{\eta}$, $\delta Q_R = Q_R\hat{\eta}_R$, $\delta Q_L = Q_L\hat{\eta}_L$, $\delta Q_A = Q_A\hat{\eta}_A$. Moreover, it is related to the angular velocity variation, for instance as $\dot{\eta} = \delta \Omega - \hat{\Omega}\eta$, implying

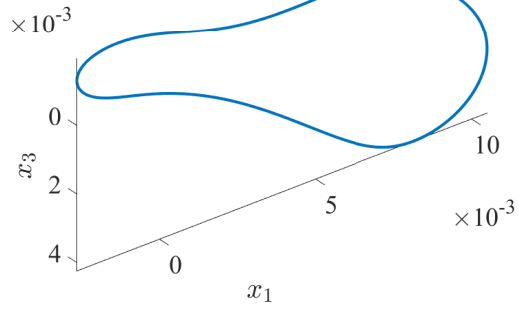
$$\delta \dot{g} = \text{ad}_{\xi}^* \cdot \delta g + \delta \xi$$

while considering δg as an element of $\mathbf{g}$.

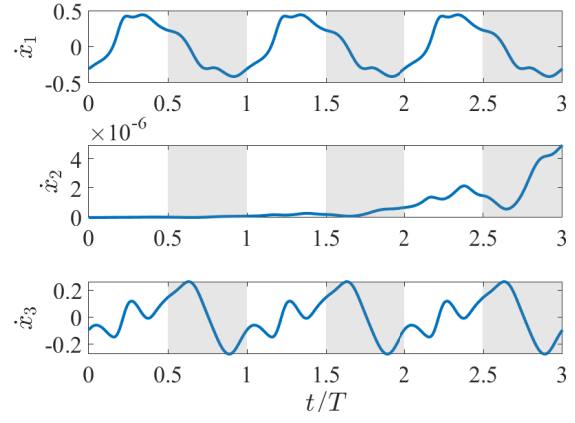
From this formulation, the linearization of dynamics in (85) will be,

$$(\delta \mathbf{J}_g) \dot{\xi} + \mathbf{J}_g(\delta \xi) - \text{ad}_{\delta \xi}^* \cdot \mathbf{J}_g(\xi) - \text{ad}_{\xi}^* \cdot \delta(\mathbf{J}_g(\xi)) + \delta(\mathbf{L}_g(\xi))(\xi) + \mathbf{L}_g(\xi)(\delta \xi) = \delta(\mathbf{f}_a + \mathbf{f}_g + \mathbf{f}_\tau)$$

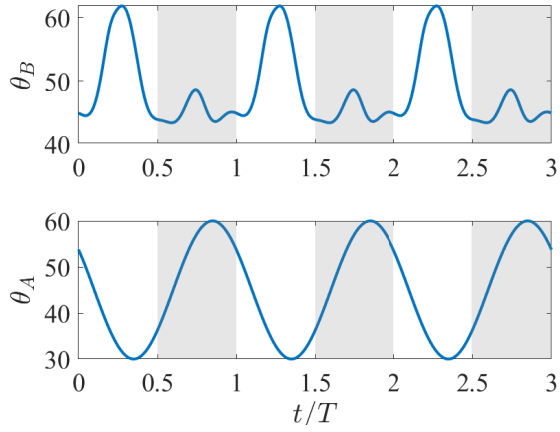
Simplifying each of the elements in this expression,



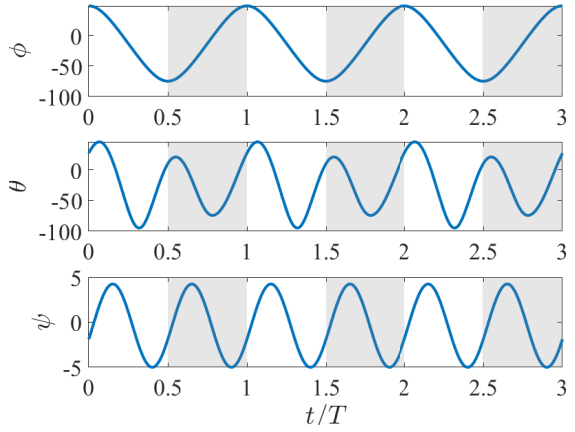
(a) Position



(b) Linear velocity



(c) Free body attitude and prescribed abdomen



(d) Free wing pitching along with prescribed flapping and deviation

Figure 13: Trajectories obtained with optimization

- $(\delta \mathbf{J}_g)\dot{\xi} = \mathbf{K}_g(\dot{\xi})(\delta g)$
 - $\text{ad}_{\delta \xi}^* \cdot \mathbf{J}_g(\xi) = -\text{ad}_{\mathbf{J}_g(\xi)}^* \cdot \delta \xi$
 - $\text{ad}_{\xi}^* \cdot \delta(\mathbf{J}_g(\xi)) = \text{ad}_{\xi}^* \cdot (\delta \mathbf{J}_g)\xi + \text{ad}_{\xi}^* \cdot \mathbf{J}_g(\delta \xi) = \text{ad}_{\xi}^* \cdot \mathbf{K}_g(\xi)(\delta g) + \text{ad}_{\xi}^* \cdot \mathbf{J}_g(\delta \xi)$
 - $\delta(\mathbf{L}_g(\xi))(\xi) = \delta(\mathbf{K}_g(\xi))(\xi) - \frac{1}{2}\delta(\mathbf{K}_g^T(\xi))(\xi)$ where, $\delta(\mathbf{K}_g(\xi))\xi = \mathbf{M}_g(\xi)(\delta g) + \mathbf{M}_{\xi}(\xi)(\delta \xi)$ and $\delta(\mathbf{K}_g^T(\xi))\xi = \tilde{\mathbf{M}}_g(\xi)(\delta g) + \tilde{\mathbf{M}}_{\xi}(\xi)(\delta \xi)$
 - $\delta(\mathbf{f}_{gravity} + \mathbf{f}_a + \mathbf{f}_{\tau}) = (\mathbf{F}_{gravity} + \mathbf{F}_a + \mathbf{F}_{\tau})(\delta g)$
- NOTE:** We have not linearized the aerodynamic forces F_R, F_L, F_A themselves

wherein,

•

$$\mathbf{M}_{gi} = \begin{bmatrix} 0 & -m_i R((\hat{\mu}_i + \widehat{Q_i \nu_i})\Omega + Q_i \hat{\nu}_i \Omega_i)^\wedge \Omega + (-\hat{\Omega} Q_i \hat{\nu}_i + Q_i \widehat{\hat{\nu}_i \Omega_i})^\wedge \Omega_i & -m_i R(\hat{\Omega}(\hat{\Omega} Q_i \hat{\nu}_i - Q_i(\hat{\nu}_i \Omega_i)^\wedge) - \hat{\Omega} Q_i(\hat{\nu}_i \Omega_i)^\wedge + Q_i((\hat{\nu}_i \Omega_i)^\wedge \Omega_i)^\wedge) \\ 0 & -m_i((\hat{\mu}_i + \widehat{Q_i \nu_i})\hat{\Omega} \widehat{R^T \dot{x}} + (Q_i \hat{\nu}_i \Omega_i)^\wedge (R^T \dot{x})^\wedge) & m_i(\widehat{R^T \dot{x}} \Omega)^\wedge Q_i \hat{\nu}_i - m_i(\widehat{R^T \dot{x}}) Q_i(\hat{\nu}_i \Omega_i)^\wedge + Q_i((J_i Q_i^T \Omega)^\wedge \Omega_i)^\wedge + Q_i \hat{\Omega}_i J_i (Q_i^T \Omega)^\wedge - Q_i(J_i \widehat{Q_i^T \Omega} \Omega_i)^\wedge - Q_i J_i \hat{\Omega}_i \widehat{Q_i^T \Omega} + m_i \hat{\mu}_i \hat{\Omega} Q_i(\hat{\nu}_i \Omega_i)^\wedge + m_i \widehat{\hat{\mu}_i \Omega} Q_i(\hat{\nu}_i \Omega_i)^\wedge + Q_i(\widehat{J_i \Omega_i} \Omega_i)^\wedge - m_i \hat{\mu}_i Q_i(\widehat{\hat{\nu}_i \Omega_i} \Omega_i)^\wedge \\ 0 & -m_i \hat{\nu}_i Q_i^T \hat{\Omega} \widehat{R^T \dot{x}} & m_i \hat{\nu}_i (Q_i^T \widehat{R^T \dot{x}} \Omega)^\wedge - m_i \hat{\nu}_i \hat{\Omega}_i (Q_i^T R^T \dot{x})^\wedge \\ & -m_i \hat{\nu}_i \hat{\Omega}_i Q_i^T (R^T \dot{x})^\wedge & -J_i \hat{\Omega}_i \widehat{Q_i^T \Omega} + m_i \hat{\nu}_i \hat{\Omega}_i (Q_i^T \hat{\mu}_i \Omega)^\wedge \end{bmatrix}$$

$$\mathbf{M}_{\xi i} = \begin{bmatrix} 0 & m_i R(-\hat{\Omega}(\hat{\mu}_i + \widehat{Q_i \nu_i}) + (Q_i \hat{\nu}_i \Omega_i)^\wedge) & -m_i R(\hat{\Omega} Q_i \hat{\nu}_i + Q_i \hat{\Omega}_i \hat{\nu}_i) \\ m_i(\hat{\mu}_i + \widehat{Q_i \nu_i})\hat{\Omega}^T R^T & Q_i \hat{\Omega}_i (J_i Q_i^T) - Q_i J_i \hat{\Omega}_i Q_i^T & Q_i \hat{\Omega}_i J_i - m_i \hat{\mu}_i Q_i \hat{\Omega}_i \hat{\nu}_i \\ -m_i (Q_i \hat{\nu}_i \Omega_i)^\wedge R^T & +m_i \hat{\mu}_i (Q_i \hat{\nu}_i \Omega_i)^\wedge + m_i (Q_i \hat{\nu}_i \Omega_i)^\wedge \hat{\mu}_i & \\ -m_i \hat{\nu}_i Q_i^T \hat{\Omega} R^T - m_i \hat{\nu}_i \hat{\Omega}_i Q_i^T R^T & -J_i \hat{\Omega}_i Q_i^T + m_i \hat{\nu}_i \hat{\Omega}_i Q_i^T \hat{\mu}_i & 0 \end{bmatrix}$$

with $\mathbf{M}_{\xi i} = \mathbf{M}_{\xi i}^T$. Hence,

$$\mathbf{M}_g(\xi) = \begin{bmatrix} 0 & \mathbf{M}_{gR12} + \mathbf{M}_{gL12} + \mathbf{M}_{gA12} & \mathbf{M}_{gR13} & \mathbf{M}_{gL13} & \mathbf{M}_{gA13} \\ 0 & \mathbf{M}_{gR22} + \mathbf{M}_{gL22} + \mathbf{M}_{gA22} & \mathbf{M}_{gR23} & \mathbf{M}_{gL23} & \mathbf{M}_{gA23} \\ 0 & \mathbf{M}_{gR32} & \mathbf{M}_{gR33} & 0 & 0 \\ 0 & \mathbf{M}_{gL32} & 0 & \mathbf{M}_{gL33} & 0 \\ 0 & \mathbf{M}_{gA32} & 0 & 0 & \mathbf{M}_{gA33} \end{bmatrix}$$

$$\mathbf{M}_{\xi}(\xi) = \begin{bmatrix} \mathbf{M}_{\xi R11} + \mathbf{M}_{\xi L11} + \mathbf{M}_{\xi A11} & \mathbf{M}_{\xi R12} + \mathbf{M}_{\xi L12} + \mathbf{M}_{\xi A12} & \mathbf{M}_{\xi R13} & \mathbf{M}_{\xi L13} & \mathbf{M}_{\xi A13} \\ & \mathbf{M}_{\xi R22} + \mathbf{M}_{\xi L22} + \mathbf{M}_{\xi A22} & \mathbf{M}_{\xi R23} & \mathbf{M}_{\xi L23} & \mathbf{M}_{\xi A23} \\ & & & 0 & 0 \\ & & & 0 & \mathbf{M}_{\xi L33} & 0 \\ & & & 0 & 0 & \mathbf{M}_{\xi A33} \end{bmatrix},$$

$$\begin{aligned}
\tilde{\mathbf{M}}_{gi} &= 2 \begin{bmatrix} 0 & 0 & 0 \\ \cdot & -m_i((\hat{\mu}_i + \widehat{Q_i \nu_i})\Omega + Q_i \hat{\nu}_i \Omega_i)^\wedge \widehat{R^T \dot{x}} & m_i \widehat{R^T \dot{x}} (\hat{\Omega} Q_i \hat{\nu}_i - Q_i (\hat{\nu}_i \Omega_i)^\wedge) \\ \cdot & \cdot & -m_i (\hat{\nu}_i (Q_i^T \hat{\Omega} R^T \dot{x})^\wedge + (\hat{\nu}_i \Omega_i)^\wedge (Q_i^T R^T \dot{x})^\wedge) \\ & & + (J_i Q_i^T \Omega)^\wedge (Q_i^T \Omega)^\wedge - (Q_i^T \Omega)^\wedge J_i (Q_i^T \Omega)^\wedge \\ & & + m_i \hat{\nu}_i (Q_i^T \hat{\Omega} \hat{\mu}_i \Omega)^\wedge + \widehat{J_i \Omega_i} \widehat{Q_i^T \Omega} \\ & & + m_i \widehat{\hat{\nu}_i \Omega_i} (Q_i^T \hat{\mu}_i \Omega)^\wedge \end{bmatrix} \\
\tilde{\mathbf{M}}_{\xi i} &= \begin{bmatrix} 0 & 0 & 0 \\ -m_i((\hat{\mu}_i + \widehat{Q_i \nu_i})\Omega)^\wedge R^T & m_i \widehat{R^T \dot{x}} (\hat{\mu}_i + \widehat{Q_i \nu_i}) & m_i \widehat{R^T \dot{x}} Q_i \hat{\nu}_i \\ -m_i (Q_i \hat{\nu}_i \Omega_i)^\wedge R^T & \cdot & \cdot \\ -m_i \hat{\nu}_i Q_i^T \hat{\Omega} R^T & m_i \hat{\nu}_i Q_i^T (R^T \dot{x})^\wedge - \widehat{Q_i^T \Omega} J_i Q_i^T & m_i (Q_i^T R^T \dot{x})^\wedge \hat{\nu}_i \\ -m_i \widehat{\hat{\nu}_i \Omega_i} Q_i^T R^T & + (J_i Q_i^T \Omega)^\wedge Q_i^T - m_i \hat{\nu}_i Q_i^T \widehat{\hat{\mu}_i \Omega} & -\widehat{Q_i^T \Omega} J_i \\ & + m_i \hat{\nu}_i Q_i^T \hat{\Omega} \hat{\mu}_i + \widehat{J_i \Omega_i} Q_i^T + m_i \widehat{\hat{\nu}_i \Omega_i} Q_i^T \hat{\mu}_i & -m_i (Q_i^T \hat{\mu}_i \Omega)^\wedge \hat{\nu}_i \end{bmatrix}
\end{aligned}$$

with $\tilde{\mathbf{M}}_{gi} = \tilde{\mathbf{M}}_{gi}^T$. Hence,

$$\begin{aligned}
\tilde{\mathbf{M}}_g(\xi) &= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ \cdot & \tilde{\mathbf{M}}_{gR_{22}} + \tilde{\mathbf{M}}_{gL_{22}} + \tilde{\mathbf{M}}_{gA_{22}} & \tilde{\mathbf{M}}_{gR_{23}} & \tilde{\mathbf{M}}_{gL_{23}} & \tilde{\mathbf{M}}_{gA_{23}} \\ \cdot & \cdot & \tilde{\mathbf{M}}_{gR_{33}} & 0 & 0 \\ \cdot & \cdot & \cdot & \tilde{\mathbf{M}}_{gL_{33}} & 0 \\ \cdot & \cdot & \cdot & \cdot & \tilde{\mathbf{M}}_{gA_{33}} \end{bmatrix} \\
\tilde{\mathbf{M}}_\xi(\xi) &= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ \tilde{\mathbf{M}}_{\xi R_{21}} + \tilde{\mathbf{M}}_{\xi L_{21}} + \tilde{\mathbf{M}}_{\xi A_{21}} & \tilde{\mathbf{M}}_{\xi R_{22}} + \tilde{\mathbf{M}}_{\xi L_{22}} + \tilde{\mathbf{M}}_{\xi A_{22}} & \tilde{\mathbf{M}}_{\xi R_{23}} & \tilde{\mathbf{M}}_{\xi L_{23}} & \tilde{\mathbf{M}}_{\xi A_{23}} \\ \tilde{\mathbf{M}}_{\xi R_{31}} & \tilde{\mathbf{M}}_{\xi R_{32}} & \tilde{\mathbf{M}}_{\xi R_{33}} & 0 & 0 \\ \tilde{\mathbf{M}}_{\xi L_{31}} & \tilde{\mathbf{M}}_{\xi L_{32}} & 0 & \tilde{\mathbf{M}}_{\xi L_{33}} & 0 \\ \tilde{\mathbf{M}}_{\xi A_{31}} & \tilde{\mathbf{M}}_{\xi A_{32}} & 0 & 0 & \tilde{\mathbf{M}}_{\xi A_{33}} \end{bmatrix}
\end{aligned}$$

$$\mathbf{F}_{gravity} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & \sum_i m_i g(\mu_i + Q_i \nu_i)^\wedge (R^T e_3)^\wedge & m_R g(R^T e_3)^\wedge Q_R \hat{\nu}_R & m_L g(R^T e_3)^\wedge Q_L \hat{\nu}_L & m_A g(R^T e_3)^\wedge Q_A \hat{\nu}_A \\ 0 & m_R g \hat{\nu}_R Q_R^T (R^T e_3)^\wedge & m_R g \hat{\nu}_R (Q_R^T R^T e_3)^\wedge & 0 & 0 \\ 0 & m_L g \hat{\nu}_L Q_L^T (R^T e_3)^\wedge & 0 & m_L g \hat{\nu}_L (Q_L^T R^T e_3)^\wedge & 0 \\ 0 & m_A g \hat{\nu}_A Q_A^T (R^T e_3)^\wedge & 0 & 0 & m_A g \hat{\nu}_A (Q_A^T R^T e_3)^\wedge \end{bmatrix}$$

$$\mathbf{F}_a = \begin{bmatrix} 0 & \sum_i -R(Q_i F_i)^\wedge & -R Q_R \hat{F}_R & -R Q_L \hat{F}_L & -R Q_A \hat{F}_A \\ 0 & 0 & -\hat{\mu}_R Q_R \hat{F}_R & -\hat{\mu}_L Q_L \hat{F}_L & -\hat{\mu}_A Q_A \hat{F}_A \end{bmatrix}, \mathbf{F}_\tau = \begin{bmatrix} 0_{2 \times 5} & 0 & 0 \\ (Q_R^T \tau_R)^\wedge & 0 & 0 \\ 0_{3 \times 2} & 0 & (Q_L^T \tau_L)^\wedge & 0 \\ 0 & 0 & 0 & (Q_A^T \tau_A)^\wedge \end{bmatrix}$$

So the linearization will simplify to,

$$\begin{aligned}\delta\dot{g} &= \text{ad}_\xi^* \cdot \delta g + \delta\xi \\ \delta\dot{\xi} &= \mathbf{J}_g^{-1}((- \mathbf{K}_g(\dot{\xi}) + \text{ad}_\xi^* \cdot \mathbf{K}_g(\xi) - (\mathbf{M}_g(\xi) - \frac{1}{2}\tilde{\mathbf{M}}_g(\xi)) + \mathbf{F}_{gravity} + \mathbf{F}_a + \mathbf{F}_\tau)(\delta g) + \\ &\quad (-\text{ad}_{\mathbf{J}_g(\xi)}^* + \text{ad}_\xi^* \cdot \mathbf{J}_g - (\mathbf{M}_\xi(\xi) - \frac{1}{2}\tilde{\mathbf{M}}_\xi(\xi)) - \mathbf{L}_g(\xi))(\delta\xi))\end{aligned}$$

3.7 Linearization with prescribed wing kinematics and abdomen/body attitude

For the simplified case of only position dynamics,

$$m\ddot{x} + \sum_{i \in \{R, L, A\}} \{ \mathbf{J}_{i12} \dot{\Omega} + \mathbf{J}_{i13} \dot{\Omega}_i + \mathbf{K}_{i12} \Omega + \mathbf{K}_{i13} \Omega_i \} = R \sum_{i \in \{R, L, A\}} Q_i F_i + mge_3$$

linearization will yield,

$$\begin{aligned}\frac{d(\delta x)}{dt} &= \delta \dot{x} \\ m \frac{d(\delta \dot{x})}{dt} &= R \sum_{i \in \{R, L, A\}} Q_i \delta F_i\end{aligned}\tag{96}$$

where,

$$\begin{aligned}F_R &= L_R + D_R \\ L_R &= \int_0^l \frac{1}{2} \rho C_L(\alpha(r)) \text{sgn}(e_1^T U_R(r) e_3^T U_R(r)) (e_2 \times U_R(r)) \|U_R(r)\| c(r) dr \\ D_R &= \int_0^l -\frac{1}{2} \rho C_D(\alpha(r)) \|U_R(r)\| U_R(r) c(r) dr \\ C_L(\alpha) &= 0.225 + 1.58 \sin((2.13\alpha^\circ - 7.2) \frac{\pi}{180}) \\ C_D(\alpha) &= 1.92 - 1.55 \cos((2.04\alpha^\circ - 9.82) \frac{\pi}{180}) \\ \alpha_R(r) &= \cos^{-1}(\frac{|e_1^T U_R(r)|}{\|U_R(r)\|}) \\ U_R(r) &= (I_{3 \times 3} - e_2 e_2^T) Q_R^T (R^T (\dot{x} - v_{\text{wind}}) + \Omega \times \mu_R) + r(Q_R \Omega + \Omega_R) \times e_2\end{aligned}$$

Hence,

$$\begin{aligned}\delta U_R(r) &= (I_{3 \times 3} - e_2 e_2^T) Q_R^T R^T \delta \dot{x} \\ \delta \alpha_R(r) &= \frac{-1}{\sin(\alpha_R(r))} \text{sgn}(e_1^T U_R(r)) e_1^T \left(I_{3 \times 3} - \frac{U_R(r) U_R^T(r)}{\|U_R(r)\|^2} \right) \frac{\delta U_R(r)}{\|U_R(r)\|} \\ \delta C_L(\alpha(r)) &= 1.58 \cos((2.13\alpha^\circ - 7.2) \frac{\pi}{180}) \times 2.13 \delta \alpha_R(r) \\ \delta C_D(\alpha(r)) &= 1.55 \sin((2.04\alpha^\circ - 9.82) \frac{\pi}{180}) \times 2.04 \delta \alpha_R(r)\end{aligned}$$

Substituting back these values of δF_R and δF_L ($\delta F_A = 0$) in (96), obtain the linearization of position dynamics.

4. Dynamic Simulation for Monarch Butterfly

4.1 Monarch Morphological Parameters

The morphological parameters and the mass distribution of Monarch are determined by an actual butterfly [ref]. The location of the leading edge and the trailing edge are measured along the span-wise direction, and they are fitted

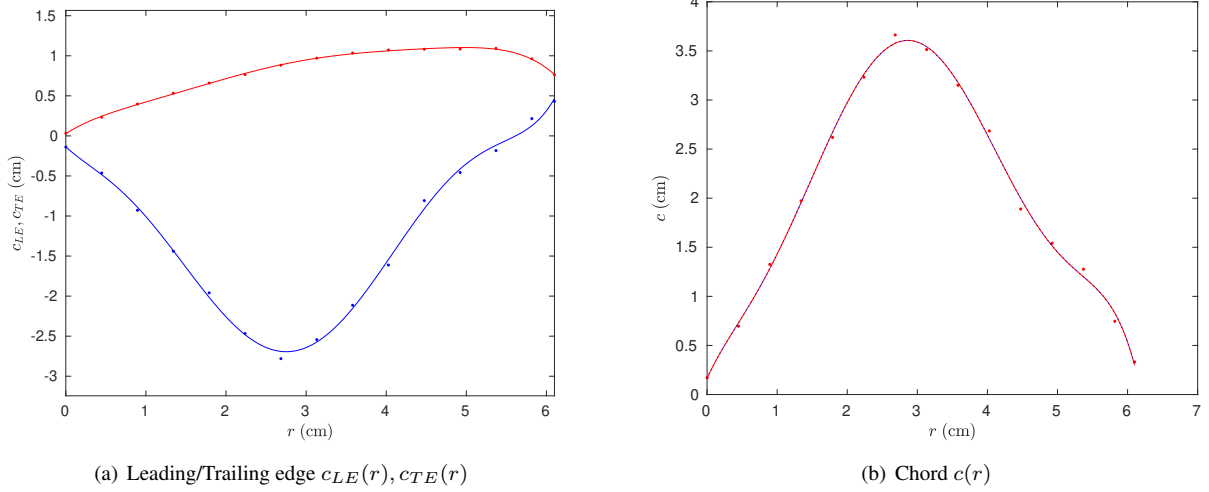


Figure 14: Monarch wing shape and chord

by sixth-order polynomials. The resulting measurements and the fitted curve are illustrated in Figure 14. From the fitted chord, we determine the morphological parameters as summarized in Table 1.

For the calculation of the inertia matrix, the head/thorax is considered as a cylinder with the height h_B and the diameter w_B . The moment of inertia of the body about its center of gravity is

$$J_B = m_B \text{diag}\left[\frac{1}{8}w_B^2, \frac{1}{16}w_B^2 + \frac{1}{12}h_B^2, \frac{1}{16}w_B^2 + \frac{1}{12}h_B^2\right].$$

The abdomen is also considered as a cylinder. Its moment of inertia about the joint is

$$J_A = m_A \text{diag}\left[\frac{1}{8}w_A^2, \frac{1}{16}w_A^2 + \frac{1}{3}h_A^2, \frac{1}{16}w_B^2 + \frac{1}{3}h_A^2\right].$$

Next, the right wing is considered as a thin plate with a uniform, negligible thickness. The mass element is written as

$$dm = \frac{m_R}{S} dx dy,$$

where $\frac{m_R}{S}$ corresponds to the area density in the unit of kg/m². Thus, the mass center of the right wing is located at

$$\begin{aligned} \nu_{R_x} &= \frac{1}{m_R} \int_{\mathcal{B}_R} x dm = \frac{1}{S} \int_0^l \int_{c_{TE}(y)}^{c_{LE}(y)} x dx dy = \frac{1}{S} \int_0^l \frac{1}{2} (c_{LE}^2(y) - c_{TE}^2(y)) dy, \\ \nu_{R_y} &= \frac{1}{m_R} \int_{\mathcal{B}_R} y dm = \frac{1}{S} \int_0^l \int_{c_{TE}(y)}^{c_{LE}(y)} y dx dy = \frac{1}{S} \int_0^l (c_{LE}(y) - c_{TE}(y)) y dy. \end{aligned}$$

Thus, the moment of inertia about the wing root is given by

$$\begin{aligned} J_{R_{xx}} &= \int_{\mathcal{B}_R} y^2 dm = \frac{m_R}{S} \int_0^l \int_{c_{TE}(y)}^{c_{LE}(y)} y^2 dx dy = \frac{m_R}{S} \int_0^l y^2 (c_{LE}(y) - c_{TE}(y)) dy, \\ J_{R_{xy}} &= \int_{\mathcal{B}_R} -xy dm = \frac{m_R}{S} \int_0^l \int_{c_{TE}(y)}^{c_{LE}(y)} -xy dx dy = \frac{m_R}{S} \int_0^l -\frac{1}{2} (c_{LE}^2(y) - c_{TE}^2(y)) y dy, \\ J_{R_{yy}} &= \int_{\mathcal{B}_R} x^2 dm = \frac{m_R}{S} \int_0^l \int_{c_{TE}(y)}^{c_{LE}(y)} x^2 dx dy = \frac{m_R}{S} \int_0^l \frac{1}{3} (c_{LE}^3(y) - c_{TE}^3(y)) dy, \\ J_{R_{zz}} &= J_{R_{xx}} + J_{R_{yy}}. \end{aligned}$$

Table 1: Monarch parameters

Head/Thorax		Abdomen		Right Wing	
m_B	$1.485 \times 10^{-4} \text{ kg}$	m_A	$1.092 \times 10^{-4} \text{ kg}$	m_R	$2.5100 \times 10^{-5} \text{ kg}$
h_B	$1.463 \times 10^{-2} \text{ m}$	h_A	$1.738 \times 10^{-2} \text{ m}$	$\bar{c}$	$2.0905 \times 10^{-2} \text{ m}$
w_B	$5.680 \times 10^{-3} \text{ m}$	w_A	$3.710 \times 10^{-3} \text{ m}$	l	$6.0996 \times 10^{-2} \text{ m}$
$J_{B_{xx}}$	$5.9887 \times 10^{-10} \text{ kgm}^2$	$J_{A_{xx}}$	$1.8788 \times 10^{-10} \text{ kgm}^2$	$J_{R_{xx}}$	$2.7568 \times 10^{-8} \text{ kgm}^2$
$J_{B_{yy}}$	$2.9481 \times 10^{-9} \text{ kgm}^2$	$J_{A_{yy}}$	$1.1089 \times 10^{-8} \text{ kgm}^2$	$J_{R_{xy}}$	$2.4957 \times 10^{-9} \text{ kgm}^2$
$J_{B_{zz}}$	$2.9481 \times 10^{-9} \text{ kgm}^2$	$J_{A_{zz}}$	$1.1089 \times 10^{-8} \text{ kgm}^2$	$J_{R_{yy}}$	$2.5799 \times 10^{-9} \text{ kgm}^2$
				$J_{R_{zz}}$	$3.0148 \times 10^{-8} \text{ kgm}^2$
				ν_{R_x}	$-4.4378 \times 10^{-3} \text{ m}$
				ν_{R_y}	$1.5176 \times 10^{-2} \text{ m}$
				S	$1.2751 \times 10^{-3} \text{ m}^2$
				$\mathcal{A}$	2.9178
				$\tilde{r}_1$	4.9761×10^{-1}
				$\tilde{r}_2$	5.4332×10^{-1}
				$\tilde{r}_3$	5.8030×10^{-1}
				r_{cp}	$6.5148 \times 10^{-2} \text{ m}$
				$\tilde{v}$	1.2496
				$\tilde{r}_1(v)$	4.8750×10^{-1}
				$\tilde{r}_2(v)$	5.1653×10^{-1}
				r_{rot}	$3.3383 \times 10^{-2} \text{ m}$

(The properties of the left wing are identical to the right wing, except $\nu_{L_y} = -\nu_{R_y}$, $J_{L_{xy}} = -J_{R_{xy}}$.)

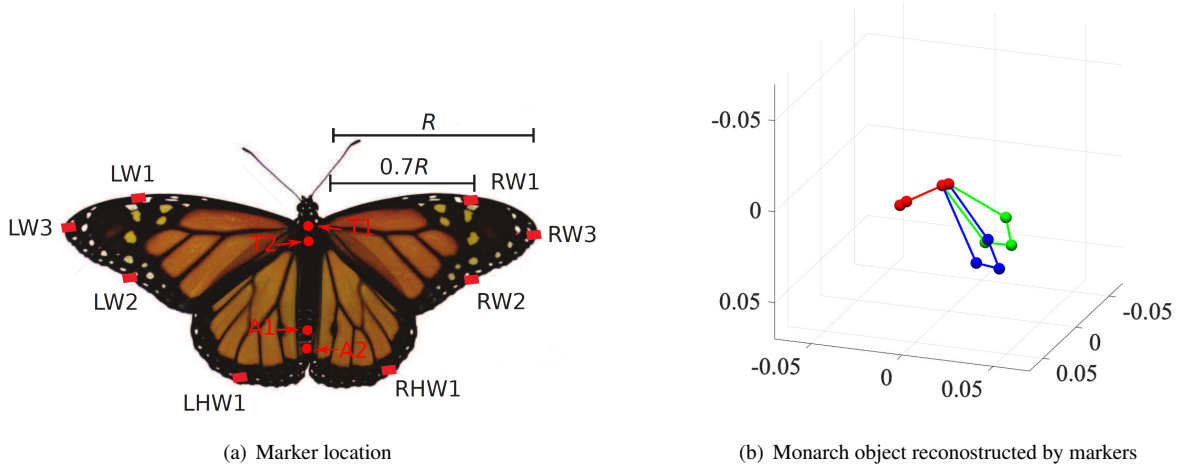


Figure 15: Monarch motion capture

Similarly, for the left wing, we have

$$\begin{aligned}
 J_{L_{xx}} &= \frac{m_R}{S} \int_0^{-l} \int_{c_{TE}(-y)}^{c_{LE}(-y)} y^2 dx dy = J_{R_{xx}} \\
 J_{L_{xy}} &= \frac{m_R}{S} \int_0^{-l} \int_{c_{TE}(-y)}^{c_{LE}(-y)} -xy dx dy = -J_{R_{xy}} \\
 J_{L_{yy}} &= \frac{m_R}{S} \int_0^{-l} \int_{c_{TE}(-y)}^{c_{LE}(-y)} x^2 dx dy = J_{R_{xx}} \\
 J_{L_{zz}} &= J_{R_{zz}}.
 \end{aligned}$$

Finally, it is assumed that the root of the wing and the joint of the abdomen are located at

$$\begin{aligned}
 \mu_R &= [0, \frac{w_B}{2}, 0], \\
 \mu_L &= [0, -\frac{w_B}{2}, 0], \\
 \mu_A &= [-\frac{h_B}{2}, 0, 0].
 \end{aligned}$$

4.2 Flight Characteristics of Monarch

The flight of an actual Monarch butterfly is studied by a motion capture system. The twelve markers are attached to a Monarch as illustrated in Figure 15, and the position of each marker is measured by a VICON motion capture system at 200Hz.

These are converted into (x, R, Q_R, Q_L, Q_A) as follows. It is considered that the origin of the body, namely x is located at the center of T_1 and T_2 . For the body attitude R , it is assumed that the first axis is along $T_1 - T_2$ and the second axis is parallel to the ground, i.e., there is no body roll. This is reasonable as the measured flight trajectory is almost straight. For the attitude of the abdomen, the first axis points from the center of A_1 and A_2 toward T_2 , and the second axis is parallel to the ground. The resulting rotation matrix is left multiplied by R^T to obtain the relative attitude Q_A .

For the right wing attitude, it is assumed that the wing root is located at the center of T_1 and T_2 . The wing plane always passes through the wing root exactly, and it is spanned by the three markers on the right wing. However, due

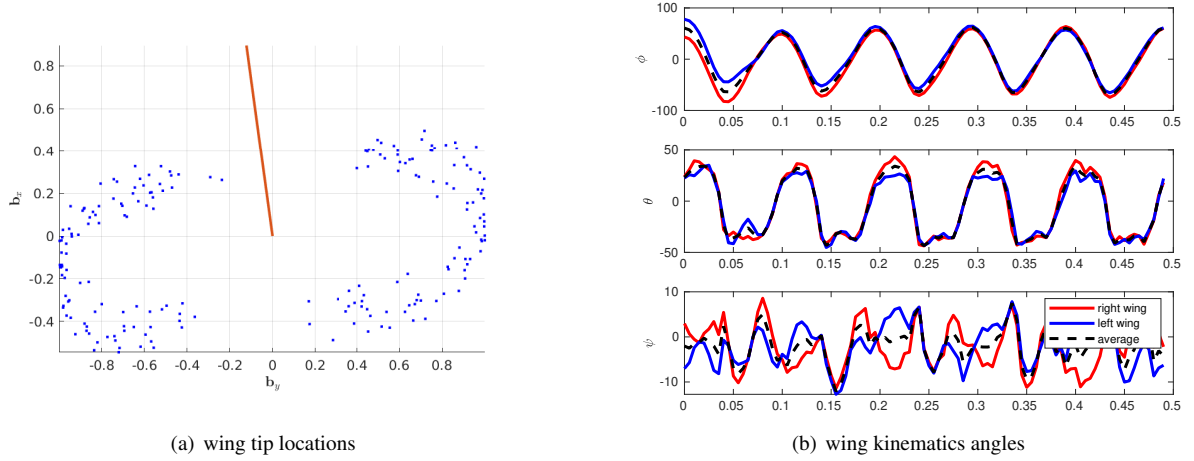


Figure 16: Monarch wing kinematics

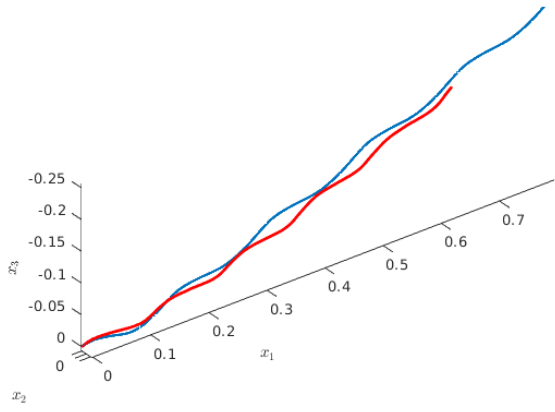
to the measurement errors and the flexibility of the wing, those points do not exactly lie on a single plane. Instead, we find the normal vector of the plane such that the sum of the squared distance between each marker and the plane is minimized. The normal vector yields the third axis of $\mathcal{F}_R$. For the second axis, the vector from the wing root to RW_3 is projected on to the fitted plane. The first axis is determined by the cross product of the second axis and the third axis. These yield the rotation matrix of the right wing from the inertial frame, and by left multiplying R^T , we obtain Q_R . The attitude of the left wing, namely Q_L is constructed similarly.

Next, the computed $(Q_R(t), Q_L(t))$ are converted into wing kinematics angles as follows. First, we determine the wing stroke plane. In the absences of the deviation angle ψ , the second column of $Q_R(t)$ and $Q_L(t)$ over multiple time instances span the stroke plane. We find the stroke plane such that the sum of the squared distance for $Q_R(t)e_2$ and $Q_L(t)e_2$ for varying t over the experiment period is minimized. It turns out that the normal vector to the stroke plane is not in the b_x - b_z plane. More explicitly, it is given by $[-0.8948, -0.1222, 0.4294]$, i.e., it is rotated by -7.77° when observed the dorsal side, or the right wing tip is ahead of the left wing tip. This might have been caused by the asymmetry of the particular Monarch butterfly used in the experiment, or the bias in the marker attachment. Instead dealing with the asymmetry in the left wing and the right wing, we multiply $\exp(7.77\hat{e}_3)$ to Q_R and Q_L such that they become symmetric in the least square sense. The resulting normal vector of the stroke plane lies in the b_x - b_z plane, and the stroke plane angle is $\beta = 25.42^\circ$. From the given β , and rotated $Q_R(t), Q_L(t)$, we can determine the wing kinematics angles $(\phi_R(t), \theta_R(t), \psi_R(t))$, and $(\phi_L(t), \theta_L(t), \psi_L(t))$ according to (3) and (6), respectively. These are illustrated in Figure 16. The wing kinematics angles for the right wing are mostly consistent with the left wing, except the small deviation angle. Assuming the symmetric wing kinematics, we take the average between the right wing and the left wing, and they are fitted with Fourier series to be used in the subsequent dynamic simulation.

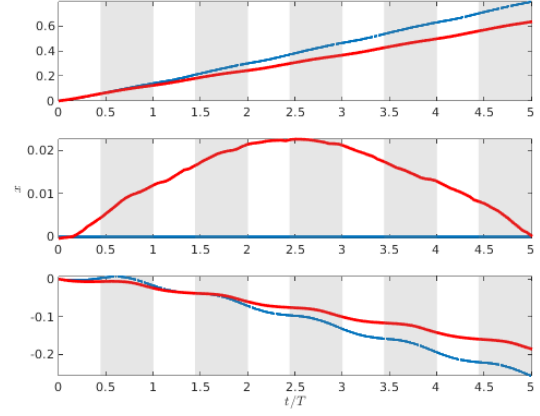
4.3 Dynamic Simulation

From the above wing kinematics angles and the body/abdomen attitude obtained by the actual Monarch butterfly, we numerically integrate the quasi-steady position dynamics, namely (95). The corresponding results are compared against the experimental data. These are illustrated at Figure 17. In general, the downstrokes generate the lift upward, and the upstrokes generate thrust forward, while yielding a climbing trajectory with oscillation.

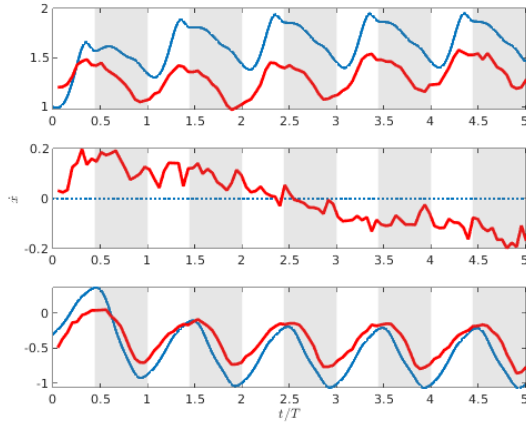
The QS model generates greater lift and thrust, and consequently causing higher climb rate and forward velocity. However, it is consistent with the experimental data in a qualitative sense.



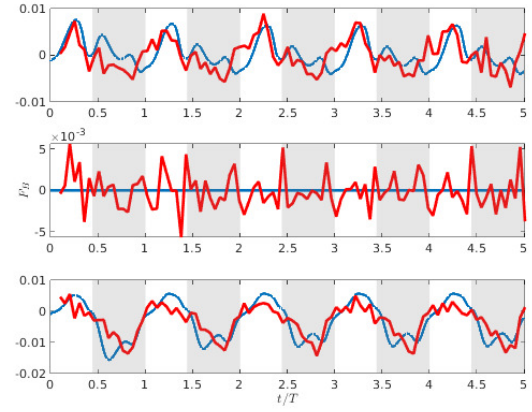
(a) Position trajectory



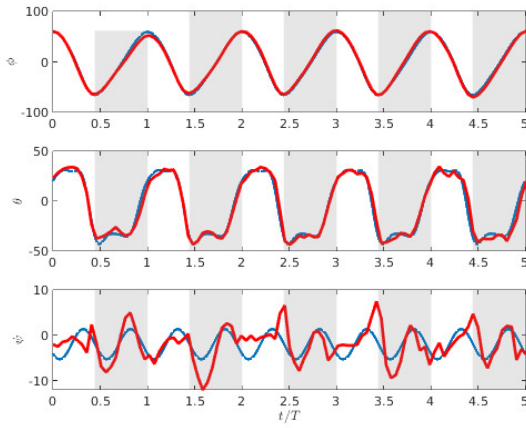
(b) Position x



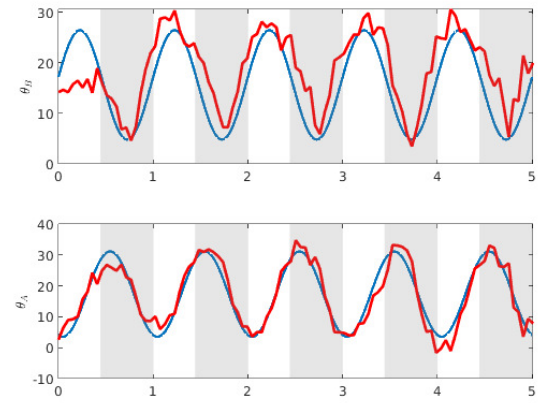
(c) Velocity $\dot{x}$



(d) Resultant force in the body-fixed frame



(e) Wing kinematics angles



(f) Body pitch / Abdomen relative pitch

Figure 17: Comparison between the QS model (blue) and the experimental data (red); shaded area corresponds to downstroke; the resultant force of the experiment is constructed by the acceleration of the thorax, multiplied by the total mass and subtracted by the gravity

4.4 Effects of Abdomen

In Monarch butterfly flights, the abdomen oscillate at the same frequency of the wing flapping angle with the opposite phase, i.e., the abdomen rotates downward during the wing upstrokes.

To test the effects of abdomen undulation with the proposed QS model, we simulate the above Monarch model for 50 strokes while fixing the abdomen pitching angle to the averaged value. These results are illustrated at Figure 18. While the discrepancy is not substantial, it is shown that the abdomen undulation slightly increases the climb rate and the forward velocity. Furthermore, an abdomen undulation in the opposite phase reduces those.

4.5 Asymptotic Stability

Another interesting results of Figure 18.(c) is that the trajectory of the velocity asymptotically converges to a periodic orbit. This suggests that the flapping motion captured by the Monarch butterfly yields an asymptotically stable periodic solution. This should be further investigated by dynamic system theory such as Floquet's theorem.

5. Floquet stability analysis

5.1 Periodic orbit generation

Ideally, we want to control the MAV so that it would be able to hover at a particular point in space. So a periodic orbit for the model is required such that at the end of a flapping cycle it comes back to its original position. This problem can be approached by using *BermanWang* model for kinematics of wing and optimizing its parameters so as to obtain hover.

Formulation of optimization:

- The objective function is directly related to absolute amount of energy into the system and the corresponding power required:

$$J(\text{parameters}) = \alpha \int_0^T \|E(t)\| dt + \beta \int_0^T \|\dot{E}(t)\| dt \quad (97)$$

wherein,

$$E(t) = \frac{1}{2}m \|\dot{x}(t)\|^2 - mge_3^T x(t), \quad \dot{E}(t) = \frac{dE(t)}{dt}$$

- Constraints for the problem are $x(0) = x(T)$, $\dot{x}(0) = \dot{x}(T)$ and all the other parameters must be kept within their physical limits
- Optimization parameters include wing kinematics angles, stroke angle, initial velocity, body and abdomen angles, and flapping frequency

Results:

Using the global optimization algorithms in MATLAB like surrogate and multistart methods, we obtain the parameters necessary for a hover trajectory.

- Parameters are constrained within the limits mentioned in Table 2 along with these additional constraints:
 - * $|\phi_m| + |\phi_0| < \pi/2$ for practically feasible flapping
 - * $x(0) = x(T)$, $\dot{x}(0) = \dot{x}(T)$ for hovering at a position
- Utilizing the parameters being optimized,
 - * Wing kinematics and subsequently attitude are determined from the *BermanWang* model in (9),(10),(11)

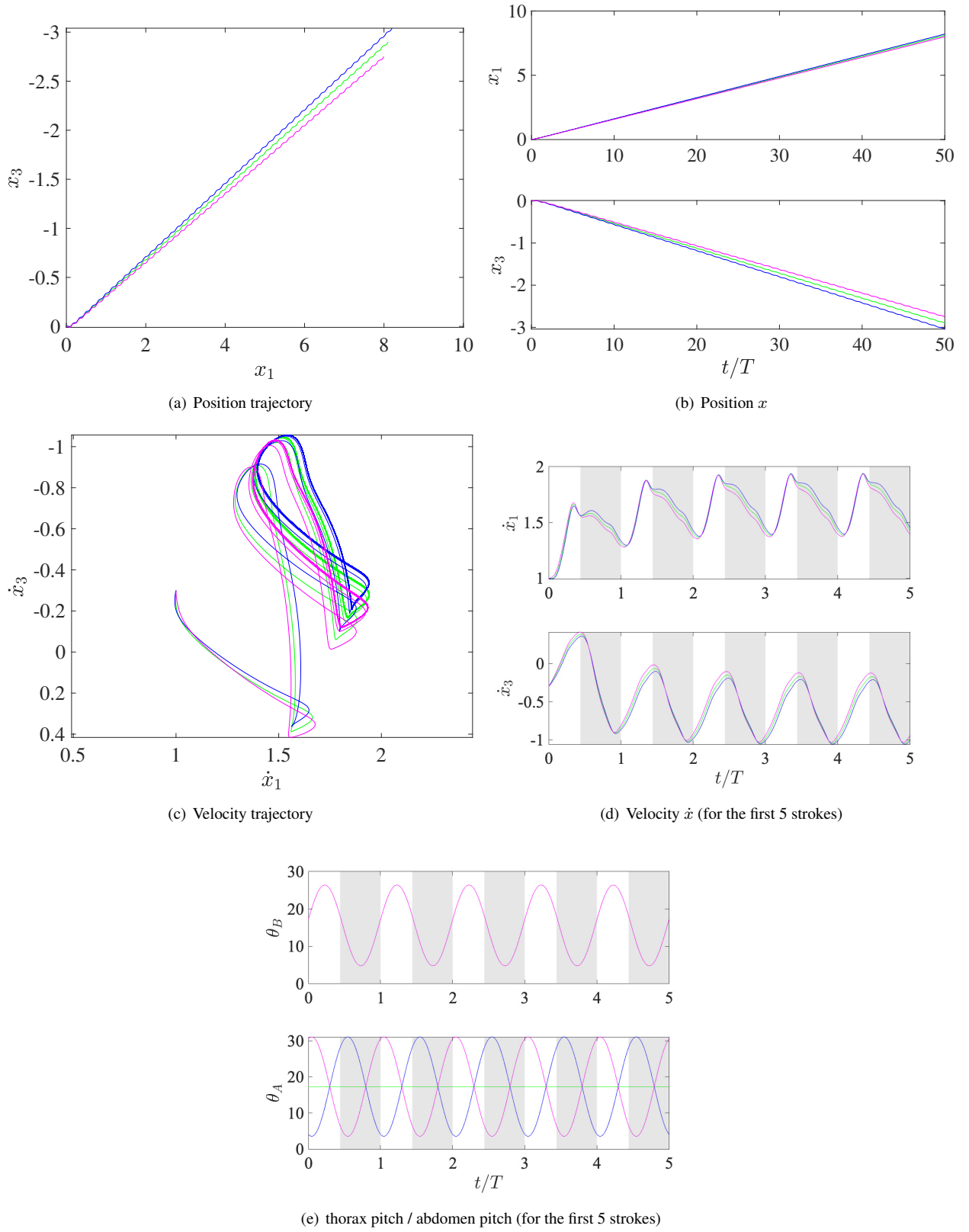


Figure 18: Comparison between flight with abdomen undulation (blue), without abdomen undulation (green), with abdomen undulation in the opposite phase (purple); the simulation is for 50 strokes, and the results of the first 5 strokes are illustrated for the subfigure (d) and (e)

- * While the body pitch is varying as $\theta_B = \theta_{B_m} \cos(2\pi ft + \theta_{B_a}) + \theta_{B_0}$, the relative abdomen pitch is either varying as $\theta_A = \theta_{A_m} \cos(2\pi ft + \theta_{A_a}) + \theta_{A_0}$ or non-oscillating in the second case as $\theta_A = \theta_{A_0}$
- * Initial velocity is considered to be $\dot{x}(0) = [\dot{x}_1(0), \dot{x}_2(0), \dot{x}_3(0)]^T$ whereas initial position is fixed at $x(0) = [0, 0, 0]^T$

Table 2: Optimized parameters

Parameters	With abdomen oscillation	Without abdomen oscillation	Lower bound	Upper bound
f	11.7584 Hz	11.7577 Hz	$(1 - 0.15)f_{natural}$	$(1 + 0.15)f_{natural}$
β	-0.2799	-0.0769	$-\pi/2$	$\pi/2$
ϕ_m	0.7425	0.7264	0	$\pi/2$
ϕ_K	0.6669	0.0894	0	1
ϕ_0	0.5665	0.7570	$-\pi/2$	$\pi/2$
θ_m	0.6981	0.6981	0	$40 \times \pi/180$
θ_C	3.0000	2.9989	0	3
θ_0	0.3914	0.3777	$-\pi/6$	$\pi/6$
θ_a	-0.2456	-0.2375	$-\pi/2$	$\pi/2$
ψ_m	2.1455×10^{-10}	0.0871	0	$8 \times \pi/180$
ψ_N	2	2	2	2
ψ_0	0.0524	0.0521	$-5 \times \pi/180$	$5 \times \pi/180$
ψ_a	-2.8879	-3.1413	$-\pi$	π
θ_{B_m}	0.1745	0.1744	0	$\pi/18$
θ_{B_0}	1.0472	0.9664	$-\pi/3$	$\pi/3$
θ_{B_a}	1.5708	1.5697	$-\pi/2$	$\pi/2$
θ_{A_m}	0.2094	—	0	$\pi/9$
θ_{A_0}	-0.1669	0.3972	$-\pi/4$	$\pi/4$
θ_{A_a}	1.5707	—	$-\pi/2$	$\pi/2$
$\dot{x}_1(0)$	-0.1000 m/s	-0.1000 m/s	-0.1 m/s	0.1 m/s
$\dot{x}_2(0)$	0.0000 m/s	0.0000 m/s	-0.1 m/s	0.1 m/s
$\dot{x}_3(0)$	-0.1000 m/s	-0.1000 m/s	-0.1 m/s	0.1 m/s
Optimized J	0.0450	0.0531	—	—

($f_{natural} = 10.2247$ Hz and ψ_N is fixed at 2)

One of the resulting solutions is illustrated in Figure ?? for which the corresponding parameters are listed in the second column of Table 2. It can be noticed that the MAV is able to come back to its initial position after the end of each flapping cycle. However, the whole body of the MAV tilts back ($\because \theta_B + \beta$ is very high) and the wings are flapping nearly horizontally so as to generate the forces necessary to counter its own weight.

To further observe the effects of abdomen, a second simulation is generated in which relative abdomen pitch is fixed at a constant value (parameters in third column of Table 2). The comparison between these two is shown in Figure ?? wherein power input at the joints are calculated as $P_R = \tau_R^T(Q_R\Omega_R)$, $P_L = \tau_L^T(Q_L\Omega_L)$, $P_A = \tau_A^T(Q_A\Omega_A)$. Relative abdomen movement (with body kept at a constant attitude) helps to decrease the amount of energy input to the MAV.

5.2 Floquet stability

For the hover trajectory in Figure ??, any disturbance in the system $(\delta x(t), \delta \dot{x}(t))$ evolves according to the dynamics in (96). It is observed that the disturbance in velocity is quickly nullified, while keeping position perturbation bounded (Figure ??).

To formalize this idea of stability, we use floquet theory [5, Section 3.5] to study the solutions of the periodic linear differential equation in (96). Consider a system of the form,

$$\dot{x} = \mathbf{A}(t)x \quad (98)$$

where $\mathbf{A}(t)$ is periodic with period T . An $n \times n$ matrix solution of this differential equation, $X(t)$, whose columns are linearly independent is called a fundamental matrix.

There exists a non-singular constant matrix $\mathbf{B}$ such that $X(t+T) = X(t)\mathbf{B}$, $\forall t$ and $\det \mathbf{B} = \exp(\int_0^T \text{tr}[\mathbf{A}(s)] ds)$. Besides, the eigenvalues $\rho_1, \dots, \rho_n$ of $\mathbf{B}$ are called characteristic multipliers and $\{\mu_j | \rho_j = e^{\mu_j T}\}$ are the corresponding characteristic exponents for the matrix differential equation. It is to be noted that ρ_j, μ_j are characteristic properties of the system (98) and are independent of the fundamental matrix. Moreover, there are n linearly independent solutions to the system which are given by,

$$\mathbf{x}_j(t) = X(t)b_j = e^{\mu_j t} \mathbf{p}_j(t)$$

where b_j are the eigenvectors of $\mathbf{B}$ and $\mathbf{p}_j(t)$ are T -periodic. If all the characteristic multipliers, $\|\rho_j\| \leq 1$, the periodic differential dynamics is stable.

For the case of disturbance dynamics of the optimized hover trajectory in Figure ??, these values turn out to be

$$\begin{aligned} \rho &= \{1.0000, 1.0000, 1.0000, 0.5951, 0.4694, 0.3697\} \\ \mu &= \{0, 0, 0, -6.0349, -8.7958, -11.5706\} \end{aligned}$$

Since all $\|\rho_j\| \leq 1$, the perturbed equation 96 is stable. Moreover the characteristic solutions, $\mathbf{x}_j(t)$, show interesting properties (non-trivial values are shown in Figure ??). The first three $j = 1, 2, 3$ correspond to constant position perturbations. However, the next $\{\mathbf{x}_j(t), j = 4, 5\}$ have zero values only along $\delta x_2, \delta \dot{x}_2$ axes (2nd, 5th), and negative characteristic exponents implying convergence of solution to zero. And finally $\mathbf{x}_6(t)$ has non-zero values only along $\delta x_2, \delta \dot{x}_2$ axes (2nd, 5th) and has the highest convergence rate. Hence, any position or velocity perturbation along the y-axis is related to an independent mode which separates it from other directions.

Furthermore, climbing trajectory of the model generated using experimental data in Figure 18 suggests that velocity values asymptotically converge to a periodic orbit. Thus we can apply floquet theory to just the velocity perturbation dynamics which is the second equation in (96). This trajectory is numerically verified to be stable since for the periodic differential equation with 3 states $(\delta \dot{x})$,

$$\begin{aligned} \rho &= \{0.2131, 0.1426, 0.2998\} \implies \|\rho_j\| \leq 1 \\ \mu &= \{-15.8085, -19.9164, -12.3171\} \end{aligned}$$

From (96), we can also conclude that the position perturbation will also be bounded. Observe that since the characteristic exponents for this trajectory are much higher in magnitude than the values for hover trajectory, it has superior stability properties.

Finally, we are interested in the stability properties of other insects as well - Hawkmoth, Bumblebee, Fruitfly. One might expect the nature of perturbation to change with flapping frequency of the wing. To begin with, the necessary morphological and aerodynamic parameters are taken from [1] for these insects while assuming the wing cross-section to be elliptical. After obtaining a simulated hover trajectory using optimization for each of these insects (see Table 3), we compare the characteristic exponents of the perturbation dynamics in Figure ?? . It is observed that, in general, the convergence rate of perturbation is positively correlated to a decrease in frequency. However, the fruitfly is an exception which shows extremely good stability characteristics.

Table 3: Optimized parameters for various insects

Parameters	Monarch	Hawkmoth	Bumblebee	Fruitfly
f	11.6286 Hz	25.2965 Hz	127.4135 Hz	259.1754 Hz
β	0.3548	0.7747	0.9117	1.4942
ϕ_m	0.6776	1.1769	1.2445	1.3162
ϕ_K	0.5100	0.7875	0.3758	0.1159
ϕ_0	0.8694	-0.3194	0.1204	0.0039
$0\theta_m$	0.6815	0.6643	0.6565	0.6929
θ_C	2.1146	1.6022	2.4667	32.6852
θ_0	0.1119	-0.0939	0.3283	-0.0298
θ_a	-0.1240	0.5546	-0.1089	-0.3111
ψ_m	0.0672	0.0177	0.0637	0.0406
ψ_N	2	2	2	2
ψ_0	0.0461	0.0366	-0.0193	0.0266
ψ_a	0.8652	0.0982	-2.2377	0.0734
$\theta_B(\text{fixed})$	0.9770	0.9012	0.0817	0.1614
θ_{A_m}	0.3277	0.1367	0.0050	0.2429
θ_{A_0}	0.4549	0.5344	-0.5937	-0.6029
θ_{A_a}	1.3473	1.5573	0.3554	-0.1486
$\dot{x}_1(0)$	-0.0600 m/s	0.0193 m/s	-0.0236 m/s	-0.0025 m/s
$\dot{x}_2(0)$	0.0000 m/s	0.0000 m/s	0.0000 m/s	0.0000 m/s
$\dot{x}_3(0)$	-0.0317 m/s	0.0376 m/s	-0.0035 m/s	-0.0326 m/s
Optimized J	0.0230	0.0092	2.7370×10^{-4}	7.6439×10^{-5}

(ψ_N is fixed at 2)

6. Control design

After studying stability of the dynamics (95), now we are going to design a suitable control so as to track the designed hover trajectory in Figure ?? . For this subsystem, we would like to design a typical PD control with the wing kinematic angles and relative abdomen attitude (pitch) as the input, and position-velocity errors as output.

Before we bring that about, we study the relation between these wing kinematic angles (from the *BermanWang* model) and the aerodynamic forces acting on the body. For instance, increasing flapping amplitude of both the wings increases force generated in the vertical direction. Figure ?? shows a detailed parametric study wherein the corresponding parameters are varied from their ideal values in the hover trajectory of Figure ?? . Here, the effect of abdomen is observed by considering $f_A = -(\mathbf{J}_{A13}\dot{\Omega}_A + \mathbf{K}_{A13}\Omega_A)$ as a contribution to the aerodynamic force in (95).

Employing these observations, a PD controller for the dynamics (95) can be designed as follows,

- Calculate the error in position and velocity as

$$\Delta x(t) = x_{des}(t) - x(t), \quad \Delta \dot{x}(t) = \dot{x}_{des}(t) - \dot{x}(t)$$

where $x_{des}, \dot{x}_{des}$ are desired position and velocity

- The error dynamics compared to the ideal version in (95) will have the form,

$$m(\ddot{x}(t) - \ddot{x}_{des}(t)) = \Delta f_a = f_a(t) - f_{a,des}(t)$$

where $f_a = RQ_R F_R + RQ_L F_L$ and assuming the angular velocities are unaffected

- The designed forces will have to be

$$\Delta f_a = m(K_P \Delta x(t) + K_D \Delta \dot{x}(t) + K_I \int_0^t \Delta x(\tau) d\tau) \quad (99)$$

where K_P, K_D, K_I are the gains

- From Figure ??, we can obtain the relation of aerodynamic forces to a few parameters which are not negligible:

$$\frac{\partial \|f_a(1)\|}{\partial \phi_m}, \quad \frac{\partial \|f_a(2)\|}{\partial \phi_m}, \quad \frac{\partial \|f_a(3)\|}{\partial \phi_m}, \quad \frac{\partial \|f_a(1)\|}{\partial \theta_m}, \quad \frac{\partial \|f_A(1)\|}{\partial \theta_{A_m}}, \quad \frac{\partial \|f_A(3)\|}{\partial \theta_{A_m}}$$

- And hence we obtain the inverse relations,

$$\begin{aligned} \Delta \theta_{A_m} &= \text{sign}(f_A(3)) \left(\frac{\partial \|f_A(3)\|}{\partial \theta_{A_m}} \right)^{-1} (w \Delta f_a(3)) \\ \Delta \phi_{m,R} &= \text{sign}(f_a(3)) \left(\frac{\partial \|f_a(3)\|}{\partial \phi_m} \right)^{-1} (1-w) \Delta f_a(3) + \text{sign}(f_a(2)) \left(\frac{\partial \|f_a(2)\|}{\partial \phi_m} \right)^{-1} \Delta f_a(2) \\ \Delta \phi_{m,L} &= \text{sign}(f_a(3)) \left(\frac{\partial \|f_a(3)\|}{\partial \phi_m} \right)^{-1} (1-w) \Delta f_a(3) - \text{sign}(f_a(2)) \left(\frac{\partial \|f_a(2)\|}{\partial \phi_m} \right)^{-1} \Delta f_a(2) \\ \Delta \theta_m &= \text{sign}(f_a(1)) \left(\frac{\partial \|f_a(1)\|}{\partial \theta_m} \right)^{-1} \left(\Delta f_a(1) - \text{sign}(f_a(1)) \frac{\partial \|f_a(1)\|}{\partial \phi_m} \frac{(\Delta \phi_{m,R} + \Delta \phi_{m,L})}{2} - \right. \\ &\quad \left. \text{sign}(f_A(1)) \frac{\partial \|f_A(1)\|}{\partial \theta_{A_m}} \Delta \theta_{A_m} \right) \end{aligned}$$

where $w \in [0, 1]$ is a weight factor deciding the contribution of abdomen

- These changes in the wing kinematic variables are further utilized to generate aerodynamic forces and thus control the system to obtain a hover trajectory:

$$\begin{aligned} m(\ddot{x}(t) - \ddot{x}_{des}(t)) &= \Delta f_a = m(K_P \Delta x(t) + K_D \Delta \dot{x}(t) + K_I \int_0^t \Delta x(\tau) d\tau) \\ \implies \Delta \ddot{x}(t) + K_D \Delta \dot{x}(t) + K_P \Delta x(t) + K_I \int_0^t \Delta x(\tau) d\tau &= 0 \end{aligned}$$

from which $\Delta x(t) \rightarrow 0$ if all the roots of $\lambda^3 + K_D \lambda^2 + K_P \lambda + K_I = 0$ have strictly negative real values

For the purposes of simulation, we consider random initial conditions for the UAV. With gain values of $K_P = 421.8868$, $K_D = 15.6030$, $K_I = 1.2655$ and corresponding roots of $-7.8 + 19i$, $-7.8 - 19i$, -0.003 , we observe in Figure ?? that the states converge to the hover trajectory.

6.1 Stability of controlled trajectory

Using the form of trajectory control in (99), we linearize the original dynamics of (95) with respect to perturbations $\delta x, \delta \dot{x}$. Denoting, $\int_0^t \delta x d\tau$ as $\delta I_x(t)$ we get,

$$\begin{aligned} \frac{d(\delta I_x)}{dt} &= \delta x \\ \frac{d(\delta x)}{dt} &= \delta \dot{x} \\ \frac{d(\delta \dot{x})}{dt} &= \frac{1}{m} R \sum_{i \in \{R, L, A\}} Q_i \delta F_i - K_P \delta \dot{x} - K_D \delta \dot{x} - K_I \delta I_x \end{aligned} \quad (100)$$

since $\delta(\Delta f_a) = m(-K_P \delta \dot{x} - K_D \delta \dot{x} - K_I \delta I_x)$

Consider this case of a periodic linear system along with the gain values used in the previous section. Here we are assuming that its trajectory has already converged to the desired periodic orbit. Now the characteristic multipliers from Floquet theory are complex numbers with the following magnitudes:

$$\|\rho\| = \{0.2592, 0.2592, 0.4139, 0.4139, 0.3591, 0.3591, 0.9997, 0.9997, 0.9997\}$$

Since all these $\|\rho_j\| \leq 1$, we can say that the perturbed dynamics in (100) is stable. Moreover the characteristic solutions (or modes), $\mathbf{x}_j(t)$, corresponding to $\rho_j \approx 1$ predominantly consist of perturbations in the integral term. Hence, we are satisfied with such values even though $\rho = 1$ implies no dissipation of perturbation.

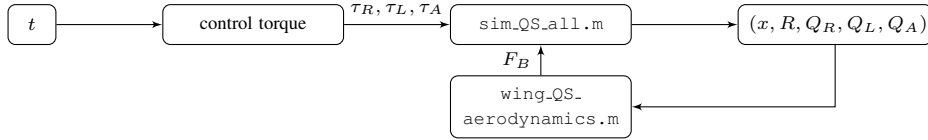
Open-Source Software Package

A software package for the proposed quasi-state dynamic model of a flapping wing UAV has been developed in Matlab, and it is shared as an open-source library at <https://github.com/fdcl-gwu/FWUAV>.

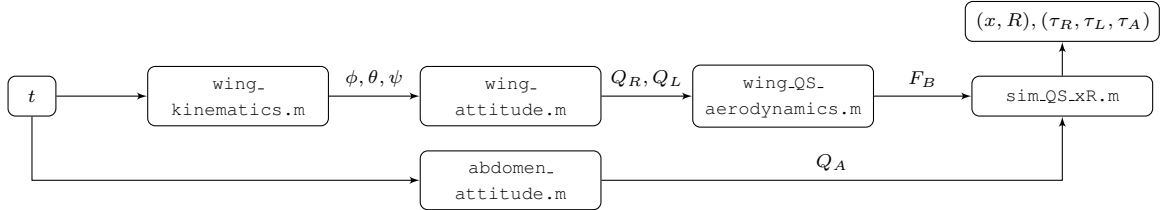
- Morphological parameters of Monarch butterfly are defied at `morp_MONARCH.m`
- The raw data from VICON are processed by the following three files under `/exp_data`.



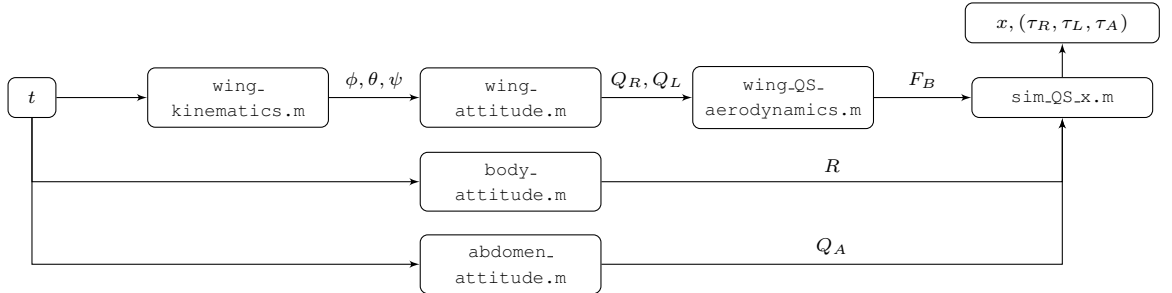
- For given torque at the wing and the abdomen, the complete dynamics (85) are simulated as follows.



- For given wing Euler angles and the abdomen attitude, the position and the attitude dynamics of the thorax, namely (90) are simulated as follows.



- For given wing Euler angles, the thorax attitude, and the abdomen attitude, the position dynamics (95) are simulated as follows.



References

- [1] G. J. Berman and Z. J. Wang, “Energy-minimizing kinematics in hovering insect flight,” *Journal of Fluid Mechanics*, vol. 582, pp. 153–168, 2007.
- [2] J. Bluman and C.-K. Kang, “Wing-wake interaction destabilizes hover equilibrium of a flapping insect-scale wing,” *Bioinspiration & biomimetics*, vol. 12, no. 4, p. 046004, 2017.
- [3] C. P. Ellington, “The aerodynamics of hovering insect flight. i. the quasi-steady analysis,” *Philosophical Transactions of the Royal Society of London. B, Biological Sciences*, vol. 305, no. 1122, pp. 1–15, 1984.
- [4] S. P. Sane and M. H. Dickinson, “The aerodynamic effects of wing rotation and a revised quasi-steady model of flapping flight,” *Journal of experimental biology*, vol. 205, no. 8, pp. 1087–1096, 2002.
- [5] G. Teschl, *Ordinary differential equations and dynamical systems*. American Mathematical Soc., 2012, vol. 140.

Appendix

Hat Map

For $x = (x_1, x_2, x_3) \in \mathbb{R}^3$,

$$\hat{x} = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}, \quad (101)$$

Sign Function

For $x \in \mathbb{R}$,

$$\text{sgn}(x) = \begin{cases} 1 & x > 0, \\ 0 & x = 0, \\ -1 & x < 0. \end{cases} \quad (102)$$

Effects of Configuration-Dependency in Inertia

$$\mathbf{L}_g = \begin{bmatrix} 0 & \mathbf{K}_{R_{12}} + \mathbf{K}_{L_{12}} + \mathbf{K}_{A_{12}} & \mathbf{K}_{R_{13}} & \mathbf{K}_{L_{13}} & \mathbf{K}_{A_{13}} \\ -\frac{1}{2}(\mathbf{K}_{R_{12}} + \mathbf{K}_{L_{12}} + \mathbf{K}_{A_{12}})^T & \mathbf{K}_{R_{22}} + \mathbf{K}_{L_{22}} + \mathbf{K}_{A_{22}} - \frac{1}{2}(\mathbf{K}_{R_{22}} + \mathbf{K}_{L_{22}} + \mathbf{K}_{A_{22}})^T & \mathbf{K}_{R_{23}} - \frac{1}{2}\mathbf{K}_{R_{32}}^T & \mathbf{K}_{L_{23}} - \frac{1}{2}\mathbf{K}_{L_{32}}^T & \mathbf{K}_{A_{23}} - \frac{1}{2}\mathbf{K}_{A_{32}}^T \\ -\frac{1}{2}\mathbf{K}_{R_{13}}^T & \mathbf{K}_{R_{32}} - \frac{1}{2}\mathbf{K}_{R_{23}}^T & \mathbf{K}_{R_{33}} - \frac{1}{2}\mathbf{K}_{R_{33}}^T & 0 & 0 \\ -\frac{1}{2}\mathbf{K}_{L_{13}}^T & \mathbf{K}_{L_{32}} - \frac{1}{2}\mathbf{K}_{L_{23}}^T & 0 & \mathbf{K}_{L_{33}} - \frac{1}{2}\mathbf{K}_{L_{33}}^T & 0 \\ -\frac{1}{2}\mathbf{K}_{A_{13}}^T & \mathbf{K}_{A_{32}} - \frac{1}{2}\mathbf{K}_{A_{23}}^T & 0 & 0 & \mathbf{K}_{A_{33}} - \frac{1}{2}\mathbf{K}_{A_{33}}^T \end{bmatrix}. \quad (103)$$